Where Does Deficit Damage Live? Actor, Network, and Replay Carriers of Post-Competence Damage in Deep Reinforcement Learning

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Abstract

In supervised deep networks, a temporary deficit early in training can cause permanent damage: a *critical period*. We ask whether temporary deficits damage deep reinforcement-learning (RL) agents beyond the training time they consume, and if so, which part of the learner's state carries the damage. We use "carries" operationally: a component carries damage if transplanting it between a damaged agent and a matched clean agent changes recovery. Every damaged run is forked from the same history as a clean run. All agents are evaluated without the deficit using common random numbers, and recovery is scored against a delay-shifted control so that lost training time is not counted as damage. State surgery then transplants the actor, critic, optimizer moments, Q-network or replay buffer. The main setting is SpaceInvaders-MinAtar with PPO, PQN and a replay-based PQN variant (PQN-R) under a temporary action restriction. Registered carrier-generalization tests add Asterix, CartPole and Acrobot, three additional impairment families, and a wider network; a Freeway reward-flip test is incomplete and descriptive. Each claim below carries its evidence strength. (i) *Timing* (exploratory, 4 seeds per learner). We saw no early critical period. Early restrictions cost only the time they occupy. Restrictions imposed after competence emerged caused *unlearning*, which was then relearned. For PPO, a pre-registered test on 12 fresh seeds *bounds* damage beyond lost time (harm larger than 3 AUC points rejected; equivalence not established) and *confirms* that restricted agents relearn faster than clean agents first learned. (ii) *Carriers* (confirmed). In PPO the recovery deficit is carried mainly by the actor's parameters (Shapley share 90%, $n = 12$). Under that movement restriction the critic adds a small, transient part, and Adam state is equivalent to zero within ± 3 AUC points. A separate onset-reference surgery localizes pause-relative harm to the actor under the SpaceInvaders reward flip (109%, $n = 8$). The actor is the largest point-estimated carrier in all eight completed settings with a detected same-update gap (82–115% shares), spanning Asterix and CartPole reward flips, a wider network, actuator noise, observation noise and reward extinction. These are registered new settings or contrasts on reused seed IDs, not eight independent new-seed replications; Acrobot is inconclusive. Freeway has only four complete seeds and is descriptive. (iii) *Replay* (bounded). In PQN-R the Q-network's parameters carry most of the damage. A ladder of increasingly matched purges shrinks the apparent effect of replay-data identity to $+0.46\,[-0.19, +1.11]$ ($n = 32$). This excludes effects of 1 AUC point or more, but not smaller ones. (iv) *Point of no return* (bounded). Two pre-registered tests detect no *average* point of no return after 2.46M restricted steps (both inconclusive; harm worse than -5 rejected at $n = 32$). One of 44 long-restricted agents (post-hoc observation; 95% interval for the rate 0.06–12%) collapsed to a near-deterministic policy and did not recover within the budget. In a pre-registered test, the policy's divergence from its clean fork at removal predicted per-agent recovery cost ($\rho = -0.48$). This is an association, with small out-of-sample gain. Our carrier results do not identify how damage is encoded, and we did not test plasticity on new tasks. All conclusions are scoped to small MLP agents, four completely tested gymnax tasks plus an incomplete fifth, and CPU-scale

budgets, and later experimental designs and primary rules were registered in a commit-timestamped log; deviations and post-hoc analyses are explicitly identified.

1 Introduction

A temporary deficit early in training can permanently impair a deep network. Achille et al. (2019) showed that blurring images for the first epochs of training leaves a lasting accuracy deficit, whereas the same blur applied later is harmless. This mirrors the critical periods of developmental neuroscience. The phenomenon has been reproduced and analysed in deep linear networks (Kleinman et al., 2024). Critical-period language has also entered RL, mostly as a question of *when* curriculum or guidance should be given (Park et al., 2021; 2022; de Kleijn et al., 2022).

RL agents face temporary deficits routinely: exploration schedules, curricula, safety constraints that disable actions, sensor failures, and non-stationary environments. RL is also unlike supervised learning in a way that matters here: the learner produces its own data. A deficit can therefore persist after it is removed in at least three places:

- in the *parameters*, as the representation damage studied under transient non-stationarity (Igl et al., 2021) and plasticity loss (Lyle et al., 2022; 2023; Dohare et al., 2024; Klein et al., 2024);
- in the *optimizer state*, whose moments can carry information across a distribution change (Dohare et al., 2023; Xu, 2026);
- in the *data*: the replay buffer, or the actor–experience loop of on-policy methods. The primacy bias (Nikishin et al., 2022) is a closely related data-and-parameter story.

These carriers call for different remedies: resets, optimizer resets, buffer purges or distillation. Yet the question of *where* damage lives is usually answered indirectly, by observing which remedy helps.

We ask three questions:

Q1. Does a temporary deficit cause damage beyond the training time it consumes, and does it depend on onset (early vs. late)?

Q2. Where is any persistent damage stored: actor, critic, optimizer moments, Q-network, replay buffer, or the data-generating loop?

Q3. Is there a restriction length beyond which damage becomes persistent (a "point of no return"), and can it be predicted at removal time?

We answer them with an interventional protocol (Figure 1). Every damaged trajectory is forked from the same training history as a matched clean trajectory. Evaluation never uses the training restriction. Recovery is compared with a delay-shifted control, so that lost training time is not counted as damage. At removal, *state surgery* composes hybrid agents whose components come from either fork, and each component's causal contribution to recovery is estimated as a factorial main effect.

Contributions.

1. **A protocol for causal localization of training damage in RL.** It combines bit-identical forks, unmasked common-random-number evaluation, delay-adjusted recovery metrics, and factorial state surgery over learner components (Section 3).
2. **An inverted timing pattern (exploratory).** In 4-seed comparisons across PPO, PQN and PQN-R, early restrictions cost only time, and late restrictions cause unlearning of acquired competence that is then largely relearned (Section 4). We detect no early critical period. For PPO, a registered test on 12 fresh seeds subsequently rejected mean harm beyond lost time worse than -3 AUC points; equivalence remained inconclusive.

3. **Carrier localization.** In PPO the same-update deficit lives mainly in the actor (exact Shapley share 90%, $n = 12$), and so does the damage beyond lost time when it exists (reward flip; surgery against the onset state). The critic adds a small, significant, transient part (12%), Adam state is equivalent to zero, and the apparent benefit of clean on-policy data is transmitted through the importance-weighted update (Section 5). In PQN-R the Q-network's parameters, not its optimizer state, carry most damage. A large buffer adds a data contribution, whose time course suggests that it acts through the network rather than as a separate store (an interpretation; Section 6). The PPO localization was tested in registered generalization settings covering four complete tasks, an incomplete fifth, five deficit families and two network widths (Section 9). The actor criterion is met in all eight completed settings with a detected same-update gap; Acrobot is unresolved and Freeway is incomplete. Independent new-seed generalization replications were planned but did not run.

4. **A replay ladder that controls the confounds.** Successively matched purge experiments, the later ones pre-registered with Holm correction and an equivalence margin, show how slot age and residence time inflate apparent data-identity effects (Section 7).

5. **Bounds on a point of no return, one failure mode, and a predictor.** A pilot and two pre-registered tests detect no average point of no return and exclude large average persistent deficits. One of 44 long-restricted agents nevertheless collapsed and never recovered. Policy divergence from the clean fork at removal is a pre-registered, Holm-supported predictor of per-agent recovery cost; three other measurements were not supported (Section 8).

How strong is each claim? The evidence is uneven across claims. Table 1 lists every headline claim with its evidence status, sample size and the wording strength we allow ourselves, and the abstract and conclusion follow it. *Confirmed*: a registered decision rule was met for a new contrast or in an explicitly reported replication. This does not automatically imply independent new seeds; Table 1 distinguishes new seed IDs from new contrasts on reused IDs. *Bounded*: a pre-registered test excluded effects beyond a stated margin but established neither presence nor equivalence. *Exploratory*: a consistent pattern in designs without a committed analysis plan, mostly with 4 seeds, which we treat as hypothesis-generating. *Post-hoc*: an observation or analysis chosen after seeing the data.

2 Related work

Critical periods. Achille et al. (2019) showed that deficits early in training permanently impair deep networks, and that the damage depends on onset and duration. Kleinman et al. (2024) found the same effect in deep linear networks and tied it to network depth and data structure. In RL, critical periods have mostly been studied as curriculum or guidance timing: when guidance is given matters for final performance (Park et al., 2021; 2022; de Kleijn et al., 2022). We instead impose a pure deficit (fewer actions) at different times and measure damage net of lost time. The answer is the opposite pattern: late, not early, deficits damage.

Plasticity loss and the primacy bias. Deep RL agents can lose the ability to learn from new data. Proposed explanations include capacity loss (Lyle et al., 2022), dormant neurons (Sokar et al., 2023; Xu et al., 2024), implicit under-parameterization (Kumar et al., 2021), and general plasticity loss under continual learning (Lyle et al., 2023; Dohare et al., 2024; Juliani & Ash, 2024); Klein et al. (2024) survey this work. Nikishin et al. (2022) showed that agents overfit early experience. Periodically resetting the last layers while keeping the buffer restores performance, resetting the optimizer alone has no effect, and resetting the buffer prevents recovery. Igl et al. (2021) showed that transient non-stationarity of targets damages generalization, and that distilling into a fresh network removes the damage. Warm-starting can also hurt generalization (Ash & Adams, 2020), and connections between plasticity and grokking have been drawn (Lyle et al., 2025). Our protocol tests which stored component these accounts implicate, by intervention; it does not test their internal mechanisms. In our setting, resetting the damaged actor is strongly harmful, and the damaged network keeps a decodable copy of the competent policy. The damage we measure therefore does not *require* plasticity loss as an explanation, although we did not measure plasticity on new tasks.

Table 1: Claims ledger: sample size, seed independence and evidence strength. F = entirely new seed IDs; M = old data plus new-seed replication; R = registered new contrasts or settings on previously used seed IDs. R is not an independent-seed replication. Detailed analysis status: Table 2.

headline claim (scope)	n; seed status	key estimate	strength
no early critical period; early restrictions cost only time (PPO, PQN, PQN-R; SpaceInvaders, movement)	4 per learner	Section 4	Exploratory
late restriction: no damage beyond lost time (PPO, SpaceInvaders, movement)	12; F (200–211)	$E_{\text{damage}} = +1.9\,[-0.7, +4.4]$; harm > 3 rejected	Bounded
late-restricted agents relearn faster than clean agents first learned (same setting)	12; F (200–211)	$E_{\text{plast}} = +14.2^{*}$	Confirmed
PPO same-update deficit lives mainly in the actor; critic small and transient; Adam ≈ 0 (SpaceInvaders, movement)	12; M (8 new)	actor 90%; fresh-only actor -14.2^{*}, critic n.s.; Adam TOST ± 3	Confirmed
damage beyond lost time lives in the actor (PPO, SpaceInvaders, reward flip, onset reference)	8; R (new reference)	actor 109% of -19.8	Confirmed
actor localization generalizes (PPO: wider net, reward flip, Asterix flip; Phase 12 panel)	8 per setting; R	8/8 detected gaps; 82–115%; Acrobot unresolved; Freeway partial	Confirmed (registered contrasts; conditional on gap)
larger gaps at removal tend to accompany larger recovery costs (PPO settings)	–	Table 14	Post-hoc
PQN-R: parameter rescue supported; optimizer-only rescue inconclusive	12 / 8; R (new split arms)	network 84%; parameter rescue $+6.4^{*}$, optimizer $+1.3$ n.s.	Exploratory / Confirmed
replay-data identity effect is within ± 1 AUC point (PQN-R, movement restriction)	32; new identity design (0–31)	$+0.46\,[-0.19, +1.11]$, TOST $p = 0.0495$	Bounded (marginal)
no average point of no return at $d = 300$ (PPO, SpaceInvaders, movement)	8 + 32; F (100–107, 300–331)†	$L_{300} = +2.3\,[-4.6, +9.2]$	Bounded (inconclusive)
extreme policy collapse without recovery improvement	44	1/44 (95% CI 0.06–12%)	Post-hoc
policy divergence at removal predicts recovery cost	32; F (300–331)	$\rho = -0.48$, Holm $p = 0.010$	Confirmed (association)†

†Completed after an unblinded interim look; the decision survives a conservative two-look adjustment (Appendix B).

Optimizer state and data as carriers. Dohare et al. (2023) argue that Adam's moment estimates contribute to policy collapse. Xu (2026) identify the first moment as a causal carrier of a source perturbation in language models, using parameter/optimizer state transplants similar to ours. Replay composition and age affect value learning (Fedus et al., 2020). Ostrovski et al. (2021) show that a passive learner trained on an active agent's data underperforms, which separates data from the learner's own interaction. Our PPO behaviour-swap and PQN-R buffer surgeries ask the complementary question: does damaged data harm a clean learner, and does clean data rescue a damaged one?

Methodology. We follow recommendations to report interval estimates over seeds (Henderson et al., 2018; Agarwal et al., 2021). Bootstrap intervals over very few seeds are anti-conservative, so we use paired t-intervals. For multiple primary tests we use Holm (1979), and for equivalence claims two one-sided tests (TOST; Lakens, 2017). Bjorck et al. (2022) show that early failures in continuous control often never recover. We ask the analogous question of whether a late failure is ever unrecoverable.

3 Setup and protocol

3.1 Environment, deficit and agents

Environment. We use SpaceInvaders-MinAtar (Young & Tian, 2019) in the gymnax implementation (Lange, 2022), run in JAX in the PureJaxRL style (Lu et al., 2022). A pilot on Breakout-, Asterix- and Freeway-MinAtar selected SpaceInvaders because it gave smooth, seed-consistent learning curves and the most signal per CPU-second. Under PPO, Breakout was bimodal across seeds: most stayed at a plateau near 7, with rare escapes. The gymnax SpaceInvaders dynamics differ from the original MinAtar (gymnax

issue #120: the enemy-bullet hit test uses the pre-move cannon position, and the first wave is larger). We study *relative* effects of deficits within one fixed MDP, so this does not affect internal validity, but our absolute returns are not comparable with MinAtar papers.

Deficit. The deficit is a fixed *action restriction* that is identical across seeds. The primary intervention allows only {no-op, fire}, removing left and right movement. A secondary intervention removes fire instead. It is a distinct intervention, not a matched-severity replicate. During the restriction the agent can select only allowed actions: PPO masks the policy logits, and PQN restricts greedy and ϵ-random action selection, and its bootstrap maxima, to the allowed set. Observations and rewards are unchanged.

Agents.

- **PPO** (Schulman et al., 2017): separate actor and critic MLPs (2×128, ReLU, orthogonal initialization). It uses 64 environments $\times$ 128 steps (8,192 environment steps per update), 4 epochs $\times$ 8 minibatches and Adam with learning rate $5 \cdot 10^{-4}$.
- **PQN** (Gallici et al., 2025): a LayerNorm Q-network (2×128) with no target network and no replay buffer, Q(λ=0.65) targets, and 128 environments $\times$ 32 steps (4,096 steps per update). It uses RAdam (Liu et al., 2019) and ϵ-greedy exploration decayed from 1 to 0.05 over the first 1M steps.
- **PQN-R**: PQN with a FIFO replay buffer (100k or 1M transitions), 1-step targets, batch 128 and replay ratio 2. Standard PQN has no buffer, so we use PQN-R wherever buffer surgery is needed.

Full hyperparameters are in Appendix A.

3.2 Timing: matched in meaning, not in update counts

Restriction windows last 409,600 environment steps for every algorithm (50 PPO updates, 100 PQN updates). *Early* windows start at initialization. *Late* windows start once the matched control has acquired substantial competence, and are matched in what the onset means rather than in update count:

- PPO: updates $[100, 150)$; the control's evaluation return at onset is 45.4, about 40% of its final level.
- PQN and PQN-R: updates $[250, 350)$; the control returns are 46.8 (PQN) and 26.6 (PQN-R).

Recovery is indexed in environment steps; individual studies use different prespecified horizons, so raw cross-algorithm effect sizes are not pooled. Phase 12 also reports a secondary PQN-R recovery grid that exactly matches the PPO grid in environment steps. Update-to-environment-step ratios, replay ratios and exploration schedules are recorded in every run manifest.

3.3 Forks, evaluation and metrics

Matched forks. A clean run C and a restricted run D share every bit of state up to the onset w_0 (same seed, same initialization, same environment and RNG streams). They diverge only in the action restriction during $[w_0, w_1)$. On CPU, same-seed runs are bit-for-bit deterministic. A control rerun matched an independent earlier run exactly (maximum absolute difference 0.0), and available same-seed reproduction checks against earlier phases are recorded in the manifests. Those checks concern the saved learner/evaluation trajectories, not every possible natural continuation.

Evaluation. Every 10 updates we run 64 episodes with the *full* action set, clean observations, true reward and a fixed evaluation key, which gives common random numbers across arms. PPO is evaluated with its stochastic policy. PQN agents are evaluated ϵ-greedily with $\epsilon = 0.05$, and greedy evaluation is secondary. The primary metric was chosen in advance from control-run reliability: greedy evaluation was 2.4$\times$ (PQN) to 2.9$\times$ (PQN-R) noisier from checkpoint to checkpoint.

Metrics and sign convention. For an arm X and reference Y, we write $\Delta(u) = X(u) - Y(u)$ for the paired difference in evaluation return at update u. **Negative always means harm.** To separate damage from lost training time, we also use the *delay-adjusted* difference against the clean control run, shifted by the window length L:

$$\Delta^{\text{delay}}(u) \;=\; D(u) - C(u - L). \tag{1}$$

$\Delta^{\text{delay}} = 0$ is what an agent that simply paused training for L updates would show. Since C and D share history up to w_0, at removal $\Delta^{\text{delay}}(w_1) = D(w_1) - D(w_0)$ is the change in competence during the restriction. A negative value means the agent *unlearned*. Recovery AUC is the mean evaluation return over a fixed post-removal window: 150 updates for PPO surgery, and 10 to 650 updates after removal for PQN.[1]

3.4 Declared estimands

"Damage" after a temporary deficit can be defined against several counterfactuals, and they give different answers (Table 5). We therefore fix one estimand per question, for the whole paper.

- **Q-damage, damage beyond the training time consumed (primary).** The estimand is the delay-adjusted recovery AUC $E_{\text{damage}} = \overline{D(w_1 + k) - C(w_1 + k - L)}$ over $k = 10..K$ (the *pause* counterfactual). Every claim of "damage", "recovery to within lost time" or "point of no return" in this paper refers to this quantity.
- **Q-plasticity, slower learning from equal competence (secondary question).** The estimand is $E_{\text{plast}} = \overline{D(w_1 + k) - C(t^\star + k)}$, where $t^\star$ is the first update at which the clean run's 3-checkpoint moving average reaches $D(w_1)$ (the *competence-matched* counterfactual). Negative values mean slower relearning, the signature of lost plasticity or of a lasting exploration deficit.
- **Total cost (descriptive).** $E_{\text{total}} = \overline{D(w_1 + k) - C(w_1 + k)}$ (the *same-update* counterfactual) is the practitioner's total cost, lost time included. We report it but do not use it to decide whether damage exists.

Why the pause baseline is primary. Of the three, it is the only counterfactual that (a) charges the deficit for exactly the training time it occupies, which is what "beyond lost time" means, and (b) is defined without conditioning on the restricted agent's post-treatment state. The same-update baseline fails (a): it counts every lost update as damage, so even a pure pause would register as harmful. The competence-matched baseline fails (b): $t^\star$ depends on how much the agent unlearned, a mediator of the effect. It therefore answers the narrower rate question, and is never used to claim that damage is absent. The two primary estimands answer different questions, so we do not correct for multiplicity between them. Component surgery with the clean fork as reference (Sections 5–9) compares damaged and clean components at the same update. It therefore attributes the same-update deficit, lost progress included, not E_{damage}. To attribute E_{damage} itself, we also run surgery against the agent's own onset state, the pause counterfactual (Section 5, Table 7). In the surgery tables, DDD−CCC is the *same-update recovery gap* being decomposed (called "total damage" in the older analysis files). It is an attribution target, not a claim of damage beyond lost time; where both are available, we report the delay-adjusted value alongside it (Table 12). These definitions were adopted after the exploratory phases, in which the pause baseline was already our delay adjustment. We tested both estimands confirmatorily on fresh seeds, pre-registered (Section 4, E1).

3.5 State surgery

At removal we save the complete learner state of both forks and compose hybrids.

- **PPO** (2^3 factorial): actor parameters, critic parameters and Adam moments, each taken from C or D. Arms are labelled by the source of (actor, critic, optimizer); e.g. DDD is the damaged agent and CDD the damaged agent with a clean actor.

[1] The repository's recovery tables report gap = control − arm (positive = damage). Every number in this paper has been converted to the arm − reference convention.

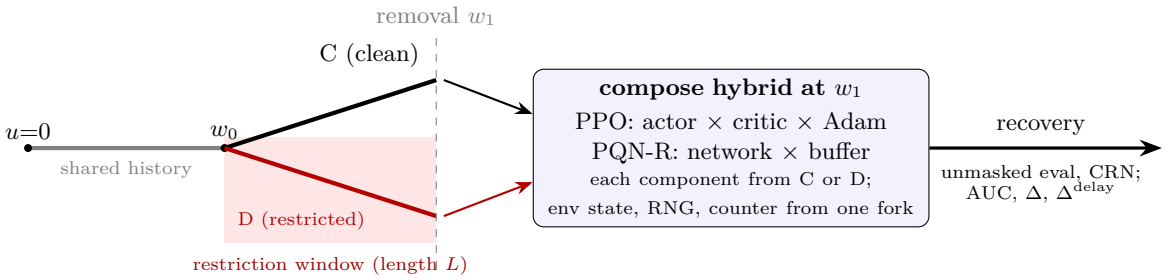


Figure 1: Protocol. A clean run C and a restricted run D share all state up to onset w_0. At removal w_1, hybrid agents are composed from components of either fork and trained without restriction. Every agent is evaluated on the full action set with common random numbers. Main effects compare the mean over hybrids holding the D version of a component with the mean over hybrids holding the C version.

- **PQN-R** (2^2): *network* (parameters together with RAdam state; this assignment was pre-specified) and *buffer* (all buffer arrays). The labels CC, DC, CD and DD give the sources of (network, buffer).

In both cases environment state, current observation, RNG and update counter come from a single fork, so the hybrids differ only in the transplanted components. A carrier's main effect is the mean over cells with its D version minus the mean over cells with its C version. We also report single transplants and rescues.

What "carrier" means in this paper (operational definition). A component is a *carrier* of damage in a given setting if transplanting it between the damaged and the clean fork at removal changes recovery. Formally, its factorial main effect, or its Shapley share of the total, differs from zero under our surgery procedure. "Damage lives in X", "X carries the damage" and "X stores the damage" are shorthand for this, and nothing more. The definition is interventional and relative to the procedure: it concerns the stored state of a component, swapped as a whole, in hybrids that do not occur in natural training (Section 5). It does *not* identify how the damage is encoded (which units, features or directions change), why the component was affected, or whether plasticity is lost. Our evidence on those questions is limited to a decodability probe and a reset arm in PPO. Learning capacity on new tasks was not measured (Section 10).

3.6 Statistics and pre-registration

All contrasts are paired by seed and reported as mean with a 95% t-interval; * marks intervals excluding 0. These are nominal, model-dependent intervals over training seeds, not distribution-free guarantees; extreme rare failures can make mean estimates particularly fragile. Seed counts are given for each experiment (4 to 32).

Our first analyses used percentile-bootstrap intervals over 4 seeds. These are anti-conservative, and all numbers reported here are the corrected t-intervals. The correction mattered once: the PPO delay-adjusted residual at +250 updates, which the bootstrap had marked as significant, is not (Section 4).

Power. Many early contrasts use $n = 4$. Large effects (the actor effect, unlearning at removal) are tight even so, but nulls at $n = 4$ are weak evidence. We handled this in three ways. (i) The two $n = 4$ nulls that carry the paper's claims were re-tested with more seeds, under a pre-registration: the PPO component surgery ($n = 12$, with equivalence tests) and the point-of-no-return test (8 fresh seeds, then 32 more). (ii) For every remaining $n = 4$ null, Appendix Table 21 reports the minimum detectable effect at 80% power and the largest harm its interval excludes. (iii) We phrase those results as bounds ("deficits worse than x are excluded"), not as absence of effect. One revision followed directly from this: with 12 seeds, a small critic effect that the 4-seed analysis missed became detectable (Section 5).

Confirmatory, exploratory and post-hoc evidence. Every result in the paper has one of three statuses (Table 2).

Table 2: Evidence status of the main claims. C = confirmatory (pre-registered before data; commit in parentheses), E = exploratory, P = post-hoc. n = seeds.

claim	section	n	status
early restrictions cost only time; late restrictions cause unlearning (PPO, PQN, PQN-R)	4	4	E
PPO late movement: E_{damage} (TOST ±3) and E_{plast}, fresh seeds	4	12	C (Phase 9, 63a8f1d)
baselines B1–B3 for the other arms and learners (Table 22)	4, App.	4	P
PPO surgery: actor, critic, optimizer main effects (Holm), TOST ±3, Shapley	5	12	C (Phase 8, 8c62fe0); seeds 0–3 E
PPO ratio-neutral behaviour swap	5	12	C (Phase 8)
PPO surgery against the onset (pause) reference, reward flip	5	8	C (Phase 10, f618d6c)
PPO surgery against the onset (pause) reference, movement	5	4	C, descriptive ($n = 4$)
surgery interaction contrasts; relative (normalized) effects	5, App.	–	P
drift suppression, actor reset, fire restriction surgery	5	4	E
PQN-R network vs. buffer surgery	6	12 / 8	E (assignment pre-specified)
PQN-R parameters vs. optimizer state	6	8	C (Phase 8)
replay ladder, rungs 1–4 (primary contrasts)	7	8–32	C (864697e, 250cec4, 2559f2e, a873dc0)
rung 1 B−E, rung 3 ANOVA, rung 4 margin sensitivity	7	8–32	P
point-of-no-return gate (pilot)	8	4	C (ba1802f)
point of no return at $d = 300$, TOST ±5	8	8	C (Phase 8)
replication at $d = 300$, TOST ±5	8	32	C (Phase 9; completed after interim look)
removal-time predictors (pilot criteria R0–R3)	8	16	C (aabd1ae)
removal-time predictors vs. A_{300} (Holm across 4)	8	32	C (Phase 9; completed after interim look)
outcome reliability; pooling across studies; seed-330 diagnosis and exclusion sensitivity	8	–	P
generalization: Asterix, reward flip, 2×256	9	8	C (Phase 8)
Asterix with the reward-flip deficit	9	8	C (Phase 9; completed after interim look)
gap at removal vs. total damage across settings (first five settings)	9	–	P
Phase 12 complete PPO panel: CartPole, Acrobot (flip); actuator noise, observation noise, extinction (SpaceInvaders)	9	8 each	C (Phase 12, adeb6b0)
Phase 12 Freeway reward flip: incomplete planned $n = 8$	9	4	E (descriptive fallback; no confirmatory verdict)
Phase 12 gap–damage prediction: no untouched completed setting in a prediction band	9	–	inconclusive (92fd75b; CartPole acute values seen)
Phase 12: PQN-R network vs. buffer under the reward flip	9	8	C (addendum bf9a946)
minimum detectable effects (Table 21)	App.	–	P

- *Confirmatory*: design, sample size, primary contrast, multiplicity rule and decision rule were committed before the test's new data were generated. Some designs reuse specified old cells or seed IDs; those are identified separately in Table 1.
- *Exploratory*: designed in advance, but without a committed analysis plan (Phases 1–5).
- *Post-hoc*: an analysis chosen after seeing the data.

Post-hoc analyses are labelled "(post-hoc)" wherever they appear. They never change or rescue a confirmatory verdict. They are reported only to qualify one (sensitivity of a margin, alternative baselines, minimum detectable effects), and every claim in the abstract and conclusion carries the strength assigned in the claims ledger (Table 1). Exploratory claims are labelled as exploratory there too. Pre-registration was introduced from the replay-purge experiments onward (Section 7). Two reviewer-response rounds (Phases 8 and 9) were pre-registered in full. Appendix B lists the commit and manifest for each result.

4 Early restrictions cost time; late restrictions cause unlearning

Evidence status. The timing comparisons in this section are **exploratory**: 4 seeds per learner, designed before our pre-registration practice and without a committed analysis plan. For PQN and PQN-R they are hypothesis-generating only. Two quantities were later **confirmed** on 12 fresh seeds for PPO's late movement restriction (Section 3.4): no damage beyond lost time larger than 3 AUC points (a bound), and faster relearning than first learning.

PPO. Table 3 and Figure 2 show the timing pattern for PPO.

Early windows. An early restriction leaves the agent far behind the control at removal ($\Delta = -22.8$), but this is entirely lost time: competence did not fall during the window ($\Delta^{\text{delay}} = +1.8$). By +250 updates the early-restricted agent is ahead of a control 50 updates younger ($\Delta^{\text{delay}} = +16.4\,[+2.1, +30.6]$). Some learning happened during the restriction.

Late windows. A late restriction causes *unlearning*: unmasked evaluation falls from 45.4 at onset to 18.5 at removal ($\Delta^{\text{delay}} = -26.9$ for movement, -30.2 for fire). It is then followed by fast relearning. After 250 updates (2.05M steps) the raw deficit (-7.2) is fully explained by lost time ($\Delta^{\text{delay}} = +2.0\,[-0.1, +4.1]$; fire $+2.4\,[-3.6, +8.3]$).

Pilots. Earlier onset sweeps with 400k-step windows at five onsets showed the same direction: later was worse at a fixed final budget. No window showed detectable damage beyond lost time. With corrected

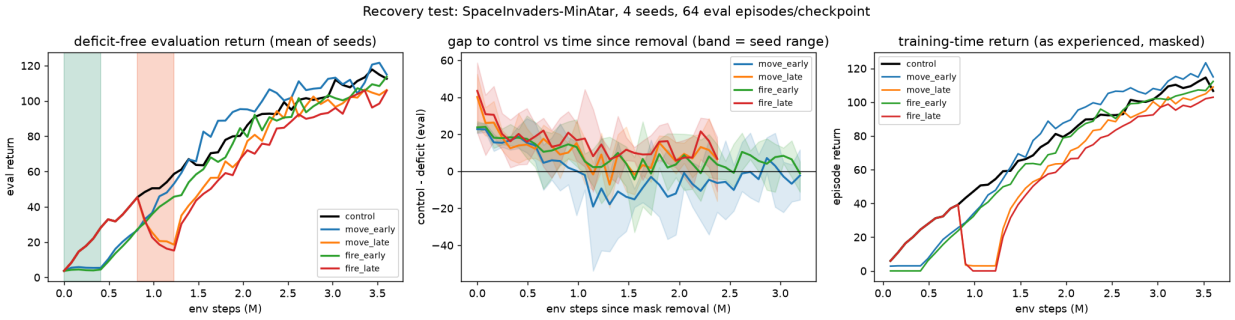


Figure 2: PPO recovery test on SpaceInvaders-MinAtar, 4 seeds, 64 evaluation episodes per checkpoint. **Left:** evaluation with the full action set (shaded: early and late windows). **Middle:** control − arm gap against time since removal (the repository's sign convention; positive = arm behind). **Right:** training return as experienced under the restriction. Early-restricted agents never had competence to lose. Late-restricted agents collapse below their competence at onset and then catch up.

Table 3: PPO recovery test ($n = 4$, paired 95% t-intervals). Δ = restricted − control at the same update; Δ^{delay} = restricted − control L=50 updates earlier. Negative = harm. At removal, Δ^{delay} is the change in competence during the window.

arm	readout	update	Δ	Δ^{delay}
move, early $[0, 50)$	at removal	50	$-22.8\,[-26.8, -18.7]^*$	$+1.8\,[+0.9, +2.6]^*$
	+50	100	$-17.0\,[-24.8, -9.1]^*$	$+0.1\,[-7.5, +7.6]$
	+250	300	$+6.6\,[-10.7, +23.9]$	$+16.4\,[+2.1, +30.6]^*$
move, late $[100, 150)$	at removal	150	$-40.1\,[-58.7, -21.6]^*$	$-26.9\,[-45.6, -8.1]^*$
	+50	200	$-13.6\,[-24.9, -2.3]^*$	$-3.3\,[-15.0, +8.5]$
	+250	400	$-7.2\,[-13.2, -1.3]^*$	$+2.0\,[-0.1, +4.1]$
fire, early $[0, 50)$	at removal	50	$-23.7\,[-28.1, -19.3]^*$	$+0.8\,[-0.2, +1.7]$
	+250	300	$-5.5\,[-18.1, +7.2]$	$+4.3\,[-19.8, +28.4]$
fire, late $[100, 150)$	at removal	150	$-43.4\,[-70.9, -15.9]^*$	$-30.2\,[-41.3, -19.1]^*$
	+250	400	$-6.9\,[-17.9, +4.1]$	$+2.4\,[-3.6, +8.3]$

t-intervals, the latest action-mask onset gave $\Delta^{\text{delay}} = -5.4\,[-11.6, +0.8]$ and observation noise at onset 0.3 gave $-2.8\,[-6.8, +1.1]$.

PQN and PQN-R. The same pattern replicates with value-based learners whose meaning-matched onset lies at a different update (Table 4; Figure 7 in the appendix).

- *Early restrictions* produce no unlearning, and the agents recover at least as fast as a delayed control: PQN's delay-adjusted AUC is $+6.4\,[+2.4, +10.5]$.
- *Late restrictions* cause severe unlearning. The restricted PQN agent's evaluation return at removal is 3.6, against 61.8 for the control, and its restriction-era policy is far below its own competence at onset.
- *After removal*, PQN recovers to within lost time (delay-adjusted AUC $-1.4\,[-3.5, +0.7]$). PQN-R is the one learner with a recovery cost beyond lost time: delay-adjusted AUC $-4.3\,[-7.0, -1.6]$. The cost is transient; by +650 updates the delay-adjusted difference is $-0.2\,[-7.7, +7.3]$.

Declared estimands on fresh seeds (confirmatory). The results above use the primary estimand, but on exploratory seeds. We therefore pre-registered a confirmatory test of both declared estimands (Section 3.4) on 12 fresh seeds (200–211), for the PPO late movement restriction with $K = 250$ (Table 5).

Table 4: PQN and PQN-R recovery (100k buffer), movement restriction ($n = 4$, $\epsilon = 0.05$ evaluation). Restriction length $L = 100$ updates (409,600 steps); AUC averages checkpoints +10 to +900 (early) or +650 (late) after removal. Negative = harm.

agent	window	readout	Δ	Δ^{delay}
PQN	early $[0, 100)$	at removal	$-19.0\,[-21.1, -16.9]^*$	$+2.4\,[-1.0, +5.9]$
		AUC	$-0.5\,[-4.5, +3.5]$	$+6.4\,[+2.4, +10.5]^*$
	late $[250, 350)$	at removal	$-58.3\,[-70.6, -45.9]^*$	$-43.2\,[-67.3, -19.2]^*$
		AUC	$-4.6\,[-7.2, -2.0]^*$	$-1.4\,[-3.5, +0.7]$
PQN-R	early $[0, 100)$	at removal	$-7.4\,[-16.4, +1.5]$	$+2.8\,[-1.6, +7.2]$
		AUC	$-2.7\,[-5.2, -0.1]^*$	$+2.7\,[-0.2, +5.5]$
	late $[250, 350)$	at removal	$-23.4\,[-32.5, -14.4]^*$	$-23.4\,[-26.8, -20.0]^*$
		+100	$-14.4\,[-22.5, -6.3]^*$	$-5.7\,[-12.0, +0.5]$
		AUC	$-8.6\,[-11.6, -5.6]^*$	$-4.3\,[-7.0, -1.6]^*$
		+650	$-3.2\,[-18.4, +12.0]$	$-0.2\,[-7.7, +7.3]$

Table 5: Confirmatory test of the declared estimands (pre-registered, Phase 9). PPO, late movement restriction $[100, 150)$, 12 fresh seeds (200–211), AUC over $k = 10..250$ after removal, paired 95% t-intervals. Negative = restricted agent worse. $t^\star$: mean 52 updates (range 30–90). Source: `p9_estimands-analysis.md`.

estimand	question	counterfactual	estimate	pre-registered decision
E_{damage} (primary)	damage beyond lost time	pause, $C(w_1 + k - L)$	$+1.9\,[-0.7, +4.4]$	inconclusive (TOST ± 3, $p = 0.17$); harm < -3 rejected, $p = 0.0007$
E_{plast}	slower relearning from equal competence	competence-matched, $C(t^\star + k)$	$+14.2\,[+10.9, +17.5]^*$	savings (12/12 seeds)
E_{total}	total cost incl. lost time	same update, $C(w_1 + k)$	$-8.9\,[-11.1, -6.6]^*$	descriptive

- **Damage beyond lost time (E_{damage}, primary).** The estimate is $+1.9\,[-0.7, +4.4]$, so the restricted agent ends slightly *ahead* of a paused control. The pre-registered TOST against ± 3 is *inconclusive* ($p = 0.17$). Its harmful side is rejected decisively: a mean below -3 is rejected at $p = 0.0007$, and the interval excludes harm beyond -0.7. Equivalence fails only because a *benefit* larger than +3 cannot be excluded. Within 250 updates, therefore, damage beyond lost time larger than about 0.7 AUC points is excluded, and the direction of any residual is open.
- **Relearning from equal competence (E_{plast}).** $+14.2\,[+10.9, +17.5]^*$, positive in 12/12 seeds. Restricted agents relearn faster than the clean agent first learned from the same competence: a savings effect, not slower relearning.
- **Total cost (E_{total}, descriptive).** $-8.9\,[-11.1, -6.6]^*$. The practitioner still pays for the lost time.

The three numbers differ in sign because they answer different questions, not because the conclusion is ambiguous. The deficit costs time ($E_{\text{total}} < 0$), it does not detectably cost more than that time (E_{damage}, with harm > 0.7 excluded), and it does not slow relearning ($E_{\text{plast}} > 0$). *Other arms and learners (post-hoc, $n = 4$).* Appendix Table 22 applies the same three counterfactuals to the exploratory runs. In every late arm the ordering is $E_{\text{total}} < E_{\text{damage}} < E_{\text{plast}}$, and E_{plast} is positive. E_{damage} is detectably negative only for PQN-R (-4.3^*), the one learner with transient damage beyond lost time. These comparisons are descriptive.

Summary. The timing pattern is the reverse of a critical period. In our data, damage beyond lost time appeared only after something had been learned that could be lost, and even then it was largely relearned. Removing movement actions after the agent has learned to dodge is what hurts.

5 In PPO, the damage lives mainly in the actor

Evidence status. The 2^3 surgery under the movement restriction and its equivalence tests are **confirmatory** ($n = 12$, 8 of them fresh seeds), and so is the onset-reference surgery under the reward flip ($n = 8$, new reference on reused seed IDs). The drift-suppression, reset and fire-restriction arms are **exploratory** ($n = 4$), and the interaction and normalized analyses are **post-hoc**.

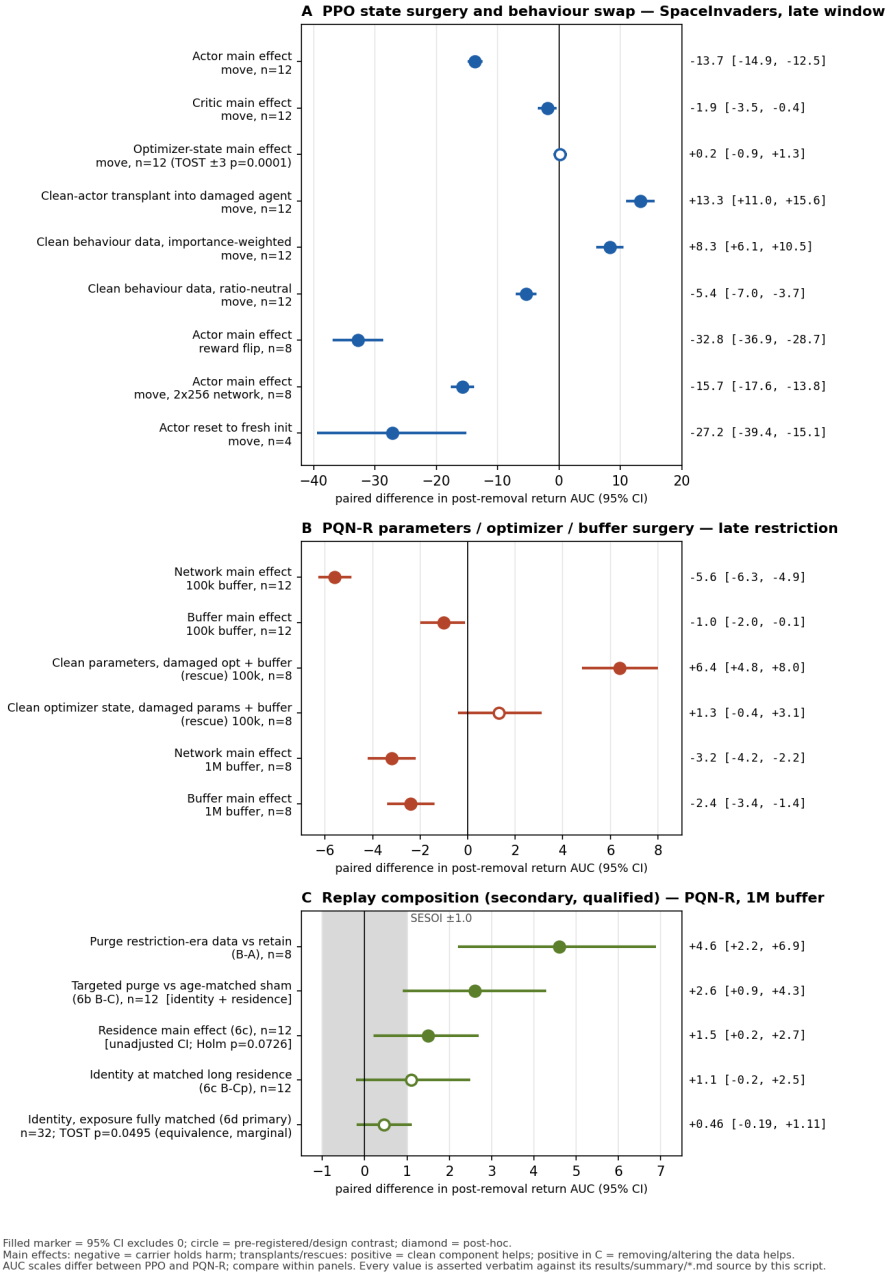


Figure 3: Summary of the causal contrasts (paired differences in post-removal return AUC, 95% t-intervals). **A**: PPO actor/critic/optimizer surgery ($n = 12$), clean-actor transplant, behaviour swap with and without importance weighting, the actor effect under the reward-flip deficit, and actor reset. **B**: PQN-R network vs. buffer surgery, and the parameter vs. optimizer-state split. **C**: the replay ladder, with the pre-registered equivalence margin ±1 (grey). Filled markers: interval excludes 0; diamond: post-hoc. For main effects, negative values mean the component holds harm; for transplants and rescues, positive values mean the clean component helps; positive values in C mean removing or altering the data helps. AUC scales differ between PPO and PQN-R; Appendix Table 23 gives the effects relative to the clean agent's AUC.

Factorial surgery. Table 6 decomposes the 150-update recovery deficit after the late movement restriction (total −15.1 AUC, $n = 12$). The first four seeds were exploratory. The eight further seeds, the multiplicity

Table 6: PPO 2^3 state surgery at removal of the late movement restriction ($n = 12$: seeds 0–3 exploratory, seeds 4–11 pre-registered replication). Recovery AUC over 150 updates. Main effects are D−C averages over the other two components (Holm across the three). Shapley values are an exact additive decomposition of the total (they sum to it per seed), and the share is the ratio of means. Arm labels give the sources of (actor, critic, optimizer). Rows marked † come from Phase 4 ($n = 4$). Source: `p8_ppo_surgery_n12-analysis.md`.

contrast	AUC	+10	Shapley value (share)
total damage DDD − CCC	−15.1 [−17.2, −13.0]*	−28.4 [−33.4, −23.4]*	−15.1 (100%)
actor main effect	−13.7 [−14.9, −12.5]*	−18.5 [−23.2, −13.9]*	−13.6 [−15.0, −12.2]* (90%)
critic main effect	−1.9 [−3.5, −0.4]*	−4.4 [−6.4, −2.5]*	−1.8 [−3.2, −0.4]* (12%)
optimizer (Adam) main effect	+0.2 [−0.9, +1.3]	−4.9 [−6.9, −3.0]*	+0.3 [−0.9, +1.5] (−2%)
actor, fresh seeds 4–11 only ($n = 8$)	−14.2 [−16.0, −12.4]*		
damaged actor only, DCC − CCC	−13.3 [−16.2, −10.4]*		
clean actor into damaged agent, CDD − DDD	+13.3 [+11.0, +15.6]*		
clean actor transplant vs. clean, CDD − CCC	−1.8 [−4.5, +0.9]		
clean critic into damaged agent, DCD − DDD	+3.0 [+0.8, +5.3]*		
actor reset to fresh init, vs. DDD†	−27.2 [−39.4, −15.1]*		
actor + optimizer reset, vs. DDD†	−24.5 [−26.8, −22.1]*		
behaviour swap for the first 10 updates ($n = 12$)			
clean behaviour, importance-weighted, DDD+bC − DDD	+8.3 [+6.1, +10.5]*	+11.8 [+7.0, +16.6]*	
clean behaviour, ratio-neutral, DDD+bCr − DDD	−5.4 [−7.0, −3.7]*	−18.0 [−22.4, −13.5]*	
damaged behaviour, importance-weighted, CCC+bD − CCC	−9.4 [−14.7, −4.2]*	−26.8 [−34.4, −19.1]*	
damaged behaviour, ratio-neutral, CCC+bDr − CCC	−2.5 [−6.6, +1.7]	−11.7 [−25.0, +1.5]	

correction, the equivalence margins and the Shapley attribution were pre-registered before any of the new data were generated.

- **Actor.** The actor main effect is −13.7 [−14.9, −12.5], and it replicates on the fresh seeds alone (−14.2 [−16.0, −12.4], $n = 8$). Main-effect "shares" of a non-additive factorial do not partition the total, so we attribute with exact Shapley values instead. These sum to the total per seed by construction. The actor's Shapley value is −13.6 [−15.0, −12.2], **90%** of the damage, negative in 12/12 seeds. A damaged actor alone in an otherwise clean agent reproduces almost all of the damage (−13.3), and a clean actor transplanted into the damaged agent rescues +13.3. After the transplant, the hybrid is statistically indistinguishable from the clean agent (−1.8 [−4.5, +0.9]).

- **Critic.** With 12 seeds the critic shows a small, detectable contribution: main effect −1.9 [−3.5, −0.4] (Holm $p = 0.038$), Shapley share 12%, clean-critic rescue +3.0 [+0.8, +5.3]. The effect is transient. It is −4.4 at +10 updates and −0.2 [−3.9, +3.6] at +150. It is not equivalent to zero within the pre-registered ±3 margin (TOST $p = 0.08$). Our 4-seed analysis had reported no detectable critic effect, which was a power limitation. *How the main effect and the Shapley value relate.* Both summarize the same four critic swaps (damaged minus clean critic, in each of the four actor × optimizer backgrounds). The main effect weights the four swaps equally. The Shapley value weights them $\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{3}$, so the all-clean and all-damaged backgrounds count more. The two coincide when there are no interactions, and here they nearly do (−1.9 vs. −1.8). Only the Shapley values add up to the total ($90\% + 12\% - 2\% = 100\%$), which is why we use them for shares. Both are significantly negative. Throughout the paper, "transient" describes the time course: the critic's contribution is concentrated in the first updates after removal and undetectable by +150. It does not mean the AUC effect is unreliable. We therefore call the critic a *minor, transient carrier*, neither uninvolved nor equivalent to zero. The same pattern recurs with the wider network (main effect −2.5, Shapley 15%; Section 9). Under the reward flip, where the actor's share is about 100%, the critic was not detectable (+1.2).

- **Optimizer state.** No damage was detected in the Adam moments, and their contribution is bounded. The main effect +0.2 [−0.9, +1.3] is statistically equivalent to zero within ±3 AUC (TOST $p = 0.0001$). A brief −4.9 dip at +10 updates, when damaged moments drive the first steps, is fully washed out by +50.

Table 7: Surgery against the primary estimand: each component's reference is the shared onset state (P, the pause counterfactual) instead of the clean fork. AUC over 150 updates; paired 95% t-intervals; Holm across the three main effects. Pre-registered (Phase 10). Sources: `p10_onsetref_rflip-analysis.md`, `p10_onsetref_move-analysis.md`, `p10_onsetref_rflip-checks.md`.

setting	n	DDD − PPP	actor	critic	Adam	actor Shapley share	PDD − DDD
SpaceInvaders, reward flip	8	$-19.8\,[-27.8, -11.8]^*$	$-22.0\,[-25.2, -18.8]^*$	$+2.8\,[-2.8, +8.4]$	$-1.6\,[-4.3, +1.0]$	109%	$+21.2\,[+14.9, +27.5]^*$
SpaceInvaders, movement	4	$-2.3\,[-4.8, +0.2]$	$-3.9\,[-5.7, -2.1]^*$	$-0.2\,[-1.2, +0.9]$	$+1.0\,[-1.9, +3.9]$	n/a (total n.s.)	$+2.7\,[-0.3, +5.8]$

- **Fire restriction.** The same holds for the distinct fire restriction ($n = 4$): total $-17.4\,[-31.2, -3.6]$, clean-actor transplant $+12.9\,[+1.3, +24.5]$.

What the surgery decomposes, and surgery against the primary estimand (pre-registered). The factorial above decomposes the *same-update* deficit DDD−CCC, which includes lost time: the clean fork's actor also holds the progress the clean agent made during the window. Our primary estimand finds no damage beyond lost time in this movement setting (Table 5). What the actor carries here is therefore mostly *lost progress*, not damage beyond lost time. To decompose the primary estimand itself, we use the fact that C and D share every bit of state up to w_0. The onset state is exactly what a paused agent would carry at removal. We repeated the surgery with each component's reference taken from the onset state (P) instead of the clean fork. The environment, observations, RNG and update counter still come from the clean fork at w_1. DDD−PPP is then damage beyond lost time, measured at the state level, and the 2^3 factorial decomposes it (Table 7).

- *Validation.* The state-level pause PPP recovers like the time-shifted control: $+1.2\,[-0.6, +3.0]$ AUC against $C(w_1 + k - L)$. DDD−PPP (-19.8) and the historical delay-adjusted AUC (-18.6) have consistent direction and magnitude, but are not bit-identical estimands: PPP uses the surgery's standardized future context. The DDD cells are bit-identical to the fork-reference run.

- *Reward flip ($n = 8$, confirmatory).* Damage beyond lost time is $-19.8\,[-27.8, -11.8]^*$. The actor main effect is $-22.0\,[-25.2, -18.8]^*$ (Holm $p < 10^{-4}$), and its Shapley value -21.6 accounts for all of it (109%; negative in 8/8 seeds). Restoring the onset actor in the damaged agent recovers $+21.2\,[+14.9, +27.5]^*$. Neither the critic ($+2.8\,[-2.8, +8.4]$) nor Adam state ($-1.6\,[-4.3, +1.0]$) contributed detectably, but neither is equivalent to zero within ± 3 at this sample size. The pre-registered localization criterion is met.

- *Movement ($n = 4$, descriptive).* Over the 150-update surgery horizon, damage beyond lost time is $-2.3\,[-4.8, +0.2]$, so by the pre-registered rule there is no damage to localize. Descriptively, the actor main effect is $-3.9\,[-5.7, -2.1]$ (Shapley -3.6, negative in 4/4 seeds), the critic and Adam effects are near zero, and restoring the onset actor recovers $+2.7\,[-0.3, +5.8]$. Any small cost beyond lost time in this setting also points to the actor, but with 4 seeds this is not a confirmed result.

So the headline claim holds for both quantities, with a qualification. Where there is damage beyond lost time (the reward flip), the actor carries it. Where there is none detectable (the movement restriction), the actor carries the lost progress that makes up the same-update deficit.

Are hybrid agents meaningful? (post-hoc checks) A hybrid such as "clean actor, damaged critic" does not arise in natural training, so a transplant effect is defined relative to the transplant procedure. Four checks bear on how much this matters (Appendix Table 24).

- *What is standardized.* CCC retains the clean learner and clean training context. DDD retains all damaged learner components but receives the same clean environment state, observation, RNG and counter as the other cells; it is not the untouched natural damaged continuation. The donor checks and repeated DDD cells establish reproducibility of this *standardized-context* procedure, not neutrality of changing the damaged agent's environment context.

- *The actor effect does not depend on the background.* Swapping in the damaged actor costs $-13.3\,[-16.2, -10.4]$ when the rest of the agent is clean, and $-13.3\,[-15.6, -11.0]$ when the rest is damaged.
- *Interactions are small.* All two- and three-way interactions are within ± 1.0 AUC points ($\leq 1.3\%$ of the clean agent's AUC), and none is significant. The total departs from the sum of the three single swaps by $-2.8\,[-5.9, +0.4]$.
- *Hybrids are not pathological.* The clean-actor hybrid CDD, whose critic and optimizer were never trained with that actor, performs like the natural clean agent ($-1.8\,[-4.5, +0.9]$).

The critic is the exception. Its swap costs nothing in an otherwise clean agent ($-0.05\,[-1.9, +1.8]$) but $-3.0\,[-5.3, -0.8]$ in an otherwise damaged one, so its small contribution is context-dependent. We therefore read the surgery as establishing causal *sufficiency within this intervention family*: putting a given component into a given agent changes recovery by a given amount. We do not read it as a description of how components interact in unperturbed training.

The damaged actor is far better than a fresh one. Re-initializing the actor while keeping the critic and optimizer is *much worse* than keeping the damaged actor: $-27.2\,[-39.4, -15.1]$ for movement and $-23.4\,[-35.6, -11.1]$ for fire. Resetting the actor's Adam moments as well changes little (-24.5).

A fresh actor in this setting learns no faster than a from-scratch agent. It reaches 26.9 after 50 updates, against 28.2 for the from-scratch control at the same point. Under this protocol the damaged actor is therefore a better starting point than a fresh one, and a reset does not help it. This is an inference about recovery on the same task. We did not probe the network's structure or its capacity to learn new tasks directly. This contrasts with the success of resets under the primacy bias (Nikishin et al., 2022).

Preventing unlearning: no recovery benefit detected ($n = 4$, exploratory). We suppressed policy drift during the restriction in two ways (Appendix Figure 9):

- F: freeze the actor's hidden layers, leaving the logit head trainable;
- A: anchor the policy to the onset policy with a KL penalty ($\beta = 1$).

Both largely prevented the collapse. Evaluation at removal improved by $+20.1\,[+8.2, +32.0]$ (F) and $+21.1\,[+13.1, +29.0]$ (A), and representation drift fell. Yet neither detectably improved recovery AUC: $+1.1\,[-3.4, +5.7]$ and $-0.0\,[-1.8, +1.7]$. Both stayed as far behind the clean agent as the unprotected one (-13.2, -14.4).

In this four-seed exploratory comparison, protecting competence did not detectably improve recovery; the intervals still allow gains up to +5.7 AUC for freezing and +1.7 for the KL anchor. What the clean actor holds that the damaged and the protected actors lack is the *progress* made during the window. This is consistent with Section 4: the persistent cost is lost time, and the actor is where most of that time's learning is stored.

The actor–experience loop. On-policy data is generated by the actor, so "data damage" in PPO cannot be separated from the actor by surgery on stored arrays alone. We swapped the behaviour policy for the first $K = 10$ post-removal updates. In the original design (exploratory, now replicated at $n = 12$), data from the clean actor rescued the damaged agent (+8.3), and data from the damaged actor harmed a clean one (-9.4). Clean data from a *different* seed had no detectable effect ($+2.4\,[-5.6, +10.4]$, $n = 4$). Neither did a tempered version of the agent's own policy, matched in KL distance from the clean policy ($-5.1\,[-14.3, +4.1]$).

That design is confounded: PPO's importance ratio π_θ/π_b now involves a foreign behaviour policy, and the clip fraction rises from 0.09 to 0.34. We therefore pre-registered *ratio-neutral* arms, in which the stored log-probability is the learner's own. PPO then sees the same states, actions and rewards from π_b, but its ratio starts at 1 and carries no importance weighting. The result is unambiguous. Without importance

Table 8: PQN-R network × buffer surgery after the late movement restriction. Network = parameters + RAdam state (separated in the text: parameters carry 87% of the network rescue). Recovery AUC over +10 to +650 updates; $\epsilon = 0.05$ evaluation. Arm labels give the sources of (network, buffer). The "gone" phase starts once no restriction-era transition remains in any buffer. †For the 100k buffer, phase contrasts use the 8 seeds (4–11) with per-slot age tags.

contrast	100k buffer ($n = 12$)	1M buffer ($n = 8$)
total damage DD − CC	$-6.7\,[-8.0, -5.3]^*$	$-5.6\,[-7.0, -4.2]^*$
network-block transfer DC − CC	$-4.6\,[-6.1, -3.0]^*$	$-2.6\,[-3.6, -1.6]^*$
data transfer CD − CC	$+0.0\,[-1.0, +1.0]$	$-1.7\,[-3.3, -0.2]^*$
network main effect	$-5.6\,[-6.3, -4.9]^*$ (84%)	$-3.2\,[-4.2, -2.2]^*$ (57%)
buffer main effect	$-1.0\,[-2.0, -0.1]^*$ (16%)	$-2.4\,[-3.4, -1.4]^*$ (43%)
interaction	$-2.1\,[-4.5, +0.2]$	$-1.3\,[-3.2, +0.6]$
restriction-era share of buffer at removal	0.96	0.41
restriction-era data gone after	+25 updates	+245 updates
buffer main effect, "gone" phase†	$-0.9\,[-2.3, +0.4]$	$-2.9\,[-4.1, -1.8]^*$
total damage, "gone" phase†	$-6.0\,[-8.1, -4.0]^*$	$-7.1\,[-9.2, -4.9]^*$
eval at removal, D vs. C network	4.2 vs. 32.0 (−87%)	14.1 vs. 27.8 (−50%)

weights, clean behaviour data *harm* the damaged learner ($-5.4\,[-7.0, -3.7]$, Holm $p < 10^{-4}$), and damaged data no longer detectably harm a clean one ($-2.5\,[-6.6, +1.7]$). The importance-weighted and ratio-neutral versions differ by $+13.6\,[+11.5, +15.7]$.

What the importance weights add is a pull toward the behaviour policy. During the swap, the KL divergence from the behaviour policy to the learner is 0.13–0.18 with importance weighting and 0.76–1.07 without it. With a clean behaviour policy, the original swap amounts to a brief off-policy distillation of the clean *actor*. The ratio-neutral clip fraction (0.20–0.23) shows that large policy changes still occur, but they come from the learner's own updates on unfamiliar data. We conclude only that the benefit of the original swap depends on importance weighting toward the clean actor. With weighting the swap acts as a brief distillation of that actor, and without it the same states, actions and rewards do not help. This is consistent with the actor-carrier account. It does not show that data content is irrelevant. Even the ratio-neutral arm is not a pure-data intervention, since it applies on-policy updates to off-policy actions, so no PPO arm cleanly isolates data content from the update.

6 In PQN-R, the network carries most of the damage

Evidence status. The network-versus-buffer surgery is **exploratory** ($n = 12$ at 100k, $n = 8$ at 1M; only the carrier assignment was pre-specified). The parameter-versus-optimizer-state split is registered ($n = 8$, new swap arms on reused histories), not an independent-seed replication. "Mediates" below is an interpretation of time courses, not a carrier measurement.

PQN-R lets us separate parameters from stored data directly. With the default 100k buffer the network dominates (Table 8): 84% of the damage, network-block transfer −4.6, data transfer $+0.0\,[-1.0, +1.0]$. The result is consistent across seeds: the network effect is negative in 12/12 seeds, with a prediction interval for a new seed of $[-8.1, -3.1]$. Restriction-era data fill 96% of the 100k buffer at removal and are fully overwritten within 25 updates. The damage that remains after that point (−6.0) is a network effect. In a 2×2 design, main effects coincide with exact Shapley values and sum to the total, so these percentages do form a partition, unlike the PPO main-effect shares in our earlier analysis.

Parameters, not optimizer state (pre-registered). Network surgery moves parameters and RAdam moments together. To separate them we ran a pre-registered 2×2 (parameters × optimizer state) at the damaged buffer ($n = 8$, 100k buffer). Its DDD cell reproduced the earlier damaged arm bit for bit in all 8 seeds. Clean parameters with the damaged optimizer state and buffer rescue $+6.4\,[+4.8, +8.0]$ AUC

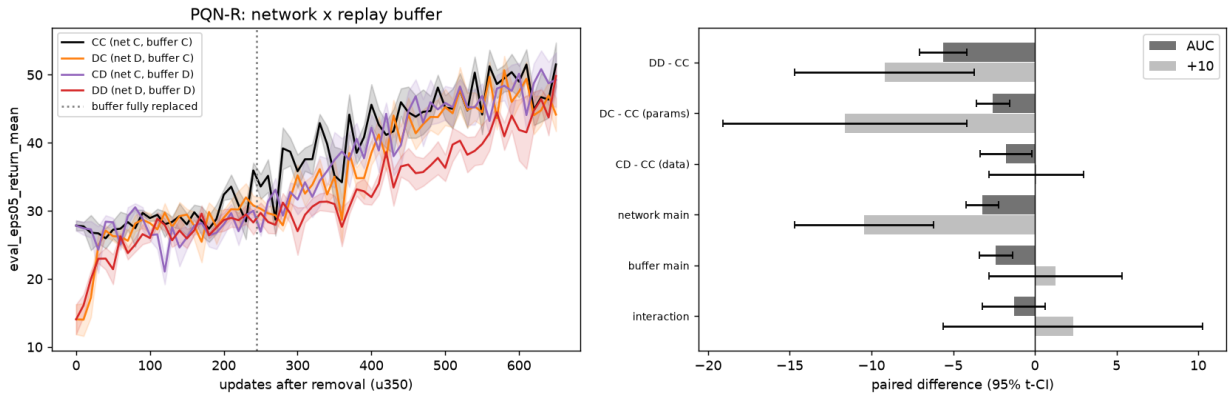


Figure 4: PQN-R network × buffer surgery with a 1M buffer ($n = 8$). **Left:** recovery curves (dotted line: the last restriction-era transition leaves the buffer at +245). **Right:** paired contrasts for AUC and for the +10 readout. A damaged network hurts recovery throughout. The damaged buffer's harm is concentrated *after* its restriction-era data is gone.

(Holm $p = 10^{-4}$), 87% of the full network rescue (+7.3). Clean optimizer state with damaged parameters rescues +1.3 [−0.4, +3.1] (Holm $p = 0.12$). The optimizer main effect in the 2×2 is small but detectable (+1.1 [+0.4, +1.9]), and it is concentrated in a transient interaction: clean parameters paired with stale RAdam moments briefly crash at +10 updates (11.1 vs. 29.9 with matched moments), then recover by +150. The PQN-R network carrier is therefore its parameters, as in PPO, where the actor parameters carry the damage and Adam state is equivalent to zero.

The buffer as a mediator. A 1M buffer is not simply a longer memory. It holds restriction-era data for 245 updates after removal (41% of slots at removal), and it also dilutes the restriction while it is happening: the collapse at removal is −50% rather than −87%.

With 1M slots the buffer main effect becomes significant (−2.4 [−3.4, −1.4], 43%). Its time course rules out the simplest reading, in which the old data is consumed and directly causes worse targets. The buffer effect is −1.4 [−2.7, −0.0] while restriction-era data are present (+10 to +150), and −2.9 [−4.1, −1.8] *after they are gone* (+250 to +650; Figure 4).

The most economical reading is that training on the restricted buffer during the first 245 updates leaves its mark in the network, and that mark outlasts the data. On this reading the buffer shapes the damage but does not store it. This is an interpretation of the time course in this PQN-R configuration (1-step targets, replay ratio 2, 1M FIFO buffer), not a direct measurement. Other buffer sizes, replay ratios or prioritized replay may behave differently.

7 A replay ladder: the data-identity effect shrinks under matching

Evidence status. All four rungs are **confirmatory** ($n = 8$ to 32, pre-registered with Holm correction). The rung-4 equivalence is marginal, so we report it as a **bound**. The ANOVA and margin-sensitivity analyses are **post-hoc**.

Is there something specific about the *identity* of restriction-era transitions, or do apparent data effects reflect how much fresh data enters the buffer and how long it stays? We ran four increasingly matched experiments, all on PQN-R with a 1M buffer. All replacement data are real transitions newly collected with a fixed protocol: a frozen clean actor (the clean fork's parameters at removal), $\epsilon = 0.05$, the full action set, fresh environments and a separate RNG stream. The pool is collected once per seed and frozen. Non-buffer state always comes from the arm's own fork. Table 9 summarizes the four experiments.

Table 9: The replay ladder (PQN-R, 1M buffer, recovery AUC over +10 to +650; positive = the first arm recovers better). Each rung matches more confounds than the one above it. The network term (bottom block) persists throughout.

	contrast	estimate	notes
1	full purge vs. retain, B − A ($n = 8$)	$+4.6\,[+2.2, +6.9]^*$	Holm $p = 0.007$
	full purge vs. purge delayed by 50 updates, B − C	$+1.0\,[-0.1, +2.1]$	Holm $p = 0.149$
2	targeted (restriction-era rows) vs. age-nearest sham, B − C ($n = 12$)	$+2.6\,[+0.9, +4.3]^*$	Holm $p = 0.006$
3	identity at matched long residence, B − Cp ($n = 12$)	$+1.1\,[-0.2, +2.5]$	Holm $p = 0.18$
	identity at matched short residence, Bp − C	$+1.1\,[-0.8, +3.0]$	Holm $p = 0.23$
	residence main effect	$+1.5\,[+0.2, +2.7]^*$	Holm $p = 0.073$
4	matched identity: R vs. U removed ($n = 32$, primary)	$+0.46\,[-0.19, +1.11]$	TOST $p = 0.0495$
	damaged vs. clean network, identical buffer (rung 2, B − D)	$-2.9\,[-4.0, -1.8]^*$	Holm $p < 0.001$
	damaged vs. clean network, identical purge (rung 1, B − E; post-hoc)	$-1.9\,[-3.0, -0.8]^*$	post-hoc

Rung 1: full purge (pre-registered, $n = 8$). Replacing the damaged agent's entire buffer at removal helps (+4.6). After the purge, recovery is indistinguishable from the clean agent with its own buffer ($-0.1\,[-2.5, +2.3]$).

The purge has a cost: a shock of $-14.8\,[-18.3, -11.3]$ at +10 updates. Its benefit is also not damage-specific. Applying the same purge to the clean agent gives $+1.8\,[-1.2, +4.8]$, and the damage-specific benefit $(B - A) - (E - D)$ is $+2.8\,[-0.2, +5.7]$. With the replacement buffer held identical, the damaged network still trails the clean one (−1.9, post-hoc).

Rung 2: targeted vs. sham (pre-registered, $n = 12$). We replaced only the restriction-era rows (3,200 rows × 128 environment columns = 409,600 transitions). The sham replaced the same number of the nearest-in-age pre-restriction rows with the same pool. Successor continuity was preserved by a per-slot seam mask.

Targeting helps (+2.6, Holm $p = 0.006$, 9/12 seeds). This contrast still confounds identity with residence: under FIFO, the sham's replacement data is overwritten about 100 updates earlier. We stated before the results that a positive B−C should not be attributed entirely to data identity.

Rung 3: identity × residence (pre-registered, $n = 12$). A 2 × 2 design crossed which block is removed with where the replacement sits. Neither identity contrast is significant (+1.1 at both residence levels). Residence has a main effect of +1.5 (Holm $p = 0.073$), and rung 2's +2.6 decomposes into +1.1 identity plus +1.5 residence.

A post-hoc repeated-measures ANOVA gives an identity main effect of $+1.11\,[+0.01, +2.21]$ ($p = 0.049$). It is concentrated in the early window ($+1.48\,[+0.79, +2.16]$).

Rung 4: matched identity (pre-registered, $n = 32$, SESOI ±1.0 AUC). *Why* ±1. The margin was fixed before the run (a873dc0) as the smallest identity effect that would justify calling restriction-era data a *practically substantial carrier* alongside the network. Smaller effects could still be carriers in the operational sense of Section 3. It is about 3% of the arms' AUC (33.1), about 30% of the targeted-purge benefit, and about the size of the rung-3 identity estimates (+1.1) at a comparable restricted exposure. For orientation, one point is 22% of the 100k-buffer network-transfer point estimate (−4.6), but 38% of the corresponding 1M-buffer estimate (−2.6, Table 8); it should not be represented as a universal one-fifth bound. The threshold is an author-chosen practical convention, not an established RL standard. It asks

whether an identity effect reaches a specified practical scale, not whether identity has any effect at all. A stricter scientific threshold would leave this experiment inconclusive. The final contrast keeps everything except identity equal. The counterpart U is the clean fork's own transitions written in the same rows, at the same updates, with the same ϵ and from the same environment instances. One common buffer per seed holds R in half of the environment columns and U in the other half; the halves were counterbalanced by seed parity.

The two arms purge either R or U with identical pool trajectories. They are matched in slot ages, residence, seams, pool exposure and replay draws (common random numbers).

- The identity effect is $+0.46\,[-0.19, +1.11]$ ($t(31) = 1.44$, $p = 0.16$, 19/32 seeds positive).
- It is statistically equivalent to zero within the pre-registered ± 1 margin, though only just (TOST $p = 0.0495$; 90% interval $[-0.08, +1.00]$).
- The pre-registered decision was "no meaningful identity effect", meaning no identity effect as large as the ± 1 margin. This is the confirmatory result.

Post-hoc sensitivity (not part of the pre-registered test; it does not alter the decision). The equivalence call is fragile: it holds for bounds of ± 1.00 and wider, fails at ± 0.75 (TOST $p = 0.18$), and needs ± 1.24 at $\alpha = 0.01$. The point estimate is positive, in the direction of the earlier rungs, and the design's minimum detectable effect (80% power) is 0.92 AUC. Our claim is therefore only that a data-identity effect of ≥ 1 AUC point (about 30% of the earlier targeted-purge benefit; these comparisons are contextual, not confidence bounds on ratios) is rejected. Smaller effects remain compatible with the data. The point estimate ($+0.46$) is positive, and we do not claim that the identity effect is zero.

Reading the ladder. As confounds are matched, the apparent harm of restriction-era data falls from $+4.6$ to $+2.6$ to $+1.1$ to $+0.46$. The network term persists at every rung. This is consistent with restriction-era data mediating damage while the network trains on them (Section 6). The ladder bounds the identity effect from above (below 1 AUC point). It does not show that the effect is zero: the point estimate is positive and in the direction of the earlier rungs. Removing restriction-era data may confer a small, early benefit beyond what any fresh, long-resident replacement data would.

8 Point of no return and removal-time predictors

Evidence status. The pilot gate ($n = 4$), the confirmatory test ($n = 8$) and the replication ($n = 32$; completed after an unblinded interim look, Appendix B) are **confirmatory**. The point-of-no-return verdicts are **bounds**, not findings of absence. The non-recovery diagnosis and the exclusion sensitivity are **post-hoc**.

Design. If damage is lost time, does it stay recoverable for arbitrarily long restrictions? All windows start at $w_0 = 100$ and are nested, so one restricted trajectory passes through the removal state for every duration d. Each removal is followed by a fixed budget of 250 recovery updates (2.05M steps).

The outcome L_d is the delay-adjusted difference averaged over the last 40 updates of the budget; A_d is the delay-adjusted AUC over the whole budget. A gate rule was committed before any data. Proceed to a confirmatory stage only if some d gives mean $L_d \leq -5$ with a 95% interval below 0; stop otherwise.

Result: STOP. No duration met the gate (Table 10), and the seed means of L_d (-0.4, $+4.8$, $+2.3$, -1.4) do not decrease with d.

- Raw damage grows almost linearly with duration, from -11.5 to -48.8. Yet a restriction three times longer than the agent's entire pre-onset history leaves an agent back at the pause-equivalent level by the end of the budget.
- The only cost beyond lost time is transient: slower early recovery at $d = 300$ ($A_{300} = -7.6\,[-10.6, -4.6]$). It did not replicate in fresh seeds (see below).

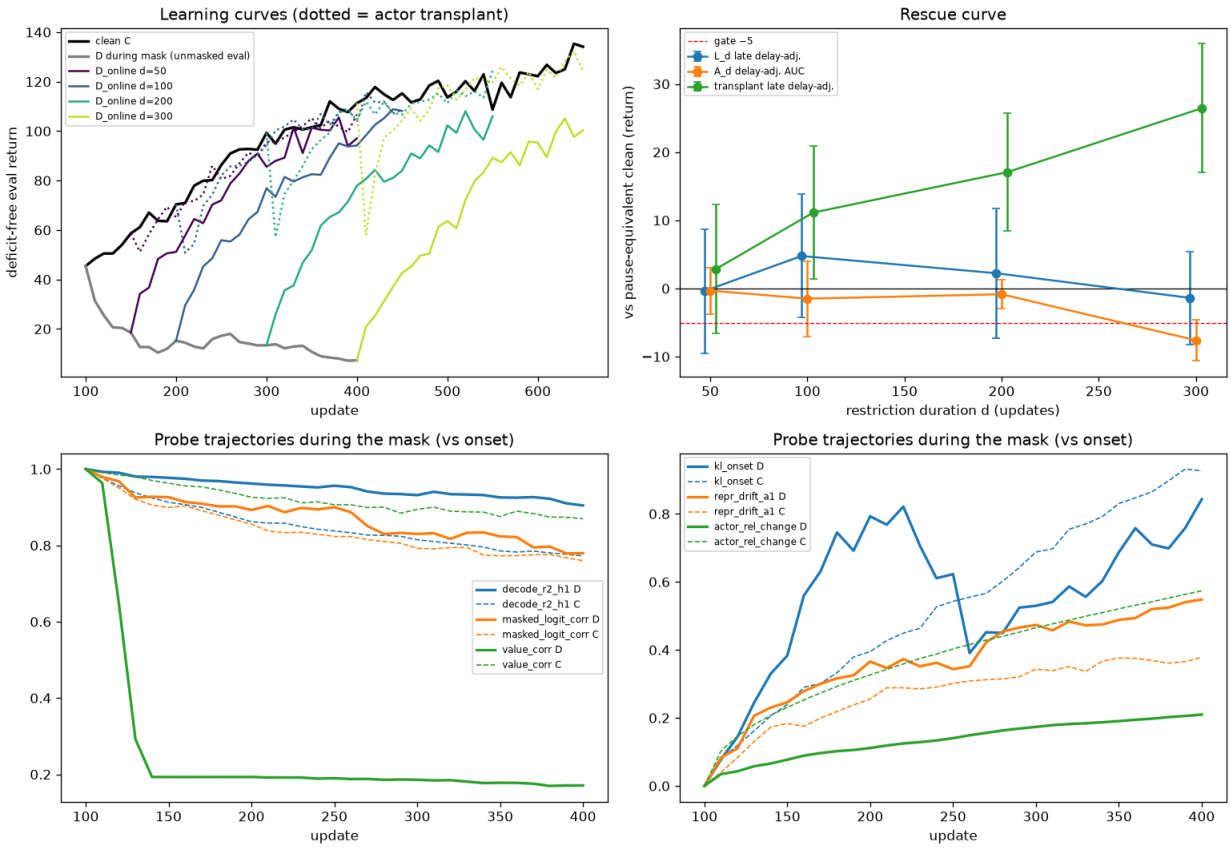


Figure 5: Point-of-no-return pilot (PPO, movement restriction from $w_0 = 100$, durations $d \in \{50, 100, 200, 300\}$, $n = 4$). **Top left:** recovery curves (dotted: clean-actor transplant). **Top right:** delay-adjusted late deficit L_d, delay-adjusted AUC A_d and transplant gain against d, with the pre-registered gate at -5. **Bottom:** probe trajectories during the restriction for D (solid) and C (dashed).

Table 10: Point-of-no-return pilot (PPO, $n = 4$). L_d: delay-adjusted difference at the end of a 250-update budget; A_d: delay-adjusted AUC; "raw" compares with the clean run at the same absolute update. Negative = harm.

d	removal eval D / C	L_d	A_d	raw AUC	transplant − D	transplant − C
50	18.5 / 58.7	−0.4 [−9.5, +8.7]	−0.3 [−3.7, +3.1]	−11.5*	+8.3*	−3.2 [−6.8, +0.5]
100	15.3 / 70.4	+4.8 [−4.3, +13.8]	−1.5 [−7.0, +4.0]	−22.6*	+18.2*	−4.4 [−9.5, +0.7]
200	13.4 / 99.3	+2.3 [−7.2, +11.7]	−0.8 [−3.0, +1.3]	−33.9*	+26.7*	−7.1 [−14.2, −0.1]*
300	7.3 / 111.3	−1.4 [−8.2, +5.5]	−7.6 [−10.6, −4.6]*	−48.8*	+42.8*	−6.1 [−14.2, +2.1]

- The competent policy stays latent. At $d = 300$, the onset policy's logits are linearly decodable from the damaged actor's features with $R^2 = 0.905$. That is higher than for the clean actor at the same update, which has moved past its onset policy.

- The critic degrades within about 30 updates (value correlation with onset ≈ 0.19) and then stays flat.

With $n = 4$ the L_d intervals are about ± 9 wide, so an impairment near -8 at $d = 300$ is not excluded. "STOP" means that none was detected.

Confirmatory test at $d = 300$ (pre-registered, 8 fresh seeds). A null gate is not evidence of absence, so we pre-registered a confirmatory test at the longest duration: an equivalence test (TOST) against the gate's ±5 margin on 8 fresh seeds (100–107). The decision rules were fixed in advance: *equivalent* if both one-sided tests reject; *persistent impairment* if the mean is ≤ -5 with an interval below 0; otherwise *inconclusive*. The result is $L_{300} = +1.1\,[-6.6, +8.7]$ (TOST $p = 0.13$): **inconclusive**. The point estimate is positive and the interval excludes impairments worse than −6.6, but one seed shows a large late deficit (−17.6), and the test lacks the power to establish equivalence within ±5. Pooled with the pilot (descriptive only, $n = 12$), $L_{300} = +0.2\,[-4.7, +5.1]$. The early cost had the same sign ($A_{300} = -4.6\,[-12.3, +3.0]$; pooled with the pilot $-5.6\,[-10.4, -0.8]$) but did not replicate in the later replication seeds (below). A clean-actor transplant again restores the agent to well above the pause reference ($+26.6\,[+20.5, +32.7]$). We therefore claim only that no point of no return was *detected* up to 2.46M restricted steps, and that persistent deficits larger than about 7 return points are excluded at $d = 300$. We do not claim that the deficit is zero.

Independent replication at $d = 300$ (pre-registered, $n = 32$). In a second reviewer-response round we pre-registered an independent replication with 32 fresh seeds (300–331). It had about 84% power to declare equivalence within ±5 if the true deficit were 0, and it included a powered test of the removal-time predictors (below). Compute loss first stopped it at 16 seeds. Those were analysed and reported descriptively in an earlier draft ($L_{300} = +6.0\,[+1.9, +10.2]$, $A_{300} = +1.3\,[-0.7, +3.3]$; no predictor supported). The remaining 16 seeds were then run with identical code and configuration. Because they were completed after this unblinded interim look, we disclose it, and we apply the pre-registered decision rules only at the pre-registered $n = 32$ (Appendix B).

At $n = 32$, $L_{300} = +2.3\,[-4.6, +9.2]$ (TOST ±5: $p_{\text{lower}} = 0.019$, $p_{\text{upper}} = 0.21$). The verdict is **inconclusive**: harm worse than −5 is rejected, but neither equivalence nor a benefit above +5 can be established. The median agent ends ahead of the pause reference (+5.7), which is consistent with the small amount of learning under the restriction that the pause baseline does not credit (Section 3.4). The early-recovery cost is $A_{300} = -3.3\,[-8.4, +1.9]$ (split-half reliability 0.86). Pooled with the Phase 8 seeds (descriptive, $n = 40$), $L_{300} = +2.1\,[-3.5, +7.6]$ and $A_{300} = -3.6\,[-7.8, +0.7]$.

One agent did not recover. Seed 330 accounts for most of the spread. Its restricted agent scored exactly 3.0 at all 26 evaluations over the 250-update recovery budget ($L_{300} = -92.0$, $A_{300} = -75.2$), while its clean fork rose normally from 46 to 114. The only other seed below −10 was 316 ($L_{300} = -11.2$). Post-hoc checks found no NaNs in the parameters and no entirely dead actor units on the evaluation-state bank. These checks do not rule out every numerical issue or establish a complete mechanism. Instead, at removal the actor had collapsed to a near-deterministic policy. Its entropy on the fixed evaluation-state bank was 0.19 nats, against 0.43–0.53 for the other restricted agents in its batch and 0.55–0.59 for clean agents, with 85% of probability mass on `fire`. PPO did not escape this state within the budget. For this agent no evaluation improvement was observed within the recovery budget; we do not claim that further training could never repair it. This is the first such case in 44 agents restricted for $d = 300$ (pilot 4, Phase 8 8, replication 32). Here the counted event is the post-hoc phenotype of an almost deterministic collapsed policy with no evaluation improvement throughout the recovery window, not every agent ending below its pause reference. Other seeds show smaller residual deficits. For this phenotype, a binomial calculation gives an exact 95% interval of 0.06%–12% (Clopper–Pearson), assuming exchangeability under this protocol; one event does not establish a stable rare-event probability. Long-lived policy collapse is a candidate individual failure mode, not proof of an absorbing state or irreversible damage. Excluding seed 330 (post-hoc sensitivity, not a decision) gives $L_{300} = +5.3\,[+2.3, +8.4]$ and $A_{300} = -1.0\,[-3.1, +1.2]$.

We therefore claim three things. No *average* point of no return was detected up to 2.46M restricted steps, and average persistent deficits worse than about 5 return points are excluded at $d = 300$. Individual agents can fail to recover within the budget. We observed one extreme flat-score collapse in 44 agents; its frequency is poorly determined, and the phenotype was classified post-hoc. We do not claim that a point of no return is absent at longer restrictions or in other settings.

Table 11: Removal-time predictors vs. recovery cost A_{300} in the replication seeds ($d = 300$, $n = 32$, pre-registered; one-sided Holm-corrected p in the direction $\rho < 0$). ρ: Spearman correlation; 95% CI by Fisher z with the Bonett–Wright standard error. The pre-registered direction is $\rho < 0$. Split-half: predictor reliability across halves of the evaluation-state bank. LOO gain: $1-\mathrm{MSE}_{\mathrm{linear}}/\mathrm{MSE}_{\mathrm{mean}}$ in leave-one-seed-out validation; both models exclude the held-out seed. This is within-cohort cross-validation, not a new cohort. Source: `p9_ponr_predictors_n32-analysis.md` (interim $n = 16$: `p9_ponr_predictors-analysis.md`).

predictor	split-half	ρ with A_{300}	95% CI for ρ	Holm p	LOO gain
policy divergence (KL)	0.95	$\mathbf{-0.48}$	$[-0.72, -0.14]$	**0.010**	+5%
representation divergence (1−CKA)	0.71	−0.19	$[-0.51, +0.17]$	0.29	−2%
restricted-action change	0.96	−0.24	$[-0.55, +0.12]$	0.27	−5%
critic error	0.71	+0.03	$[-0.32, +0.38]$	0.57	−3%

Removal-time predictors (pre-specified): policy divergence supported. We then asked whether a measurement taken at removal predicts *which* agents recover slowly, beyond duration. Before computing any metric we fixed four candidates and pass criteria R0–R3. The candidates were:

- policy divergence from the clean fork (KL);
- representation divergence (1−CKA; Kornblith et al., 2019);
- relative change of the restricted actions' probabilities in relevant states;
- critic error against the clean critic.

The criteria were: R0, measurement reliability; R1, pooled rank correlation with recovery cost; R2, rank correlation within duration; R3, leave-one-seed-out predictive gain over a duration-only model.

All four were reliable measurements (R0: 0.87 to 0.99) and correlated with recovery cost when pooled across durations. None passed: within-duration correlations were +0.05, −0.11, +0.13 and −0.35, and every candidate *reduced* out-of-fold accuracy (gains of −25%, −8%, −8% and −83%). The pooled associations come entirely from the shared trend with duration. One pre-specified secondary analysis, re-computing the metrics early in recovery, was not run; we disclose this as a deviation (Appendix B).

The pilot criteria compare durations. The replication instead tests, at a single duration ($d = 300$, $n = 32$), whether the same four measurements predict per-agent recovery cost A_{300}. The test is one-sided in the pre-registered direction (more divergence, slower recovery) with Holm correction across the four (Table 11). **Policy divergence from the clean fork is supported**: $\rho = -0.48\,[-0.72, -0.14]$, Holm $p = 0.010$. The other three are not: 1−CKA −0.19, restricted-action change −0.24 (Holm $p = 0.27$ and 0.29), critic error +0.03. The non-recovering seed 330 had the largest divergence of all 32 agents. The rank association remains after excluding seed 330 (post-hoc): $\rho = -0.43\,[-0.69, -0.07]$ (one-sided $p = 0.0078$, Holm $p = 0.031$). Its strength is not insensitive to all choices: combining that exclusion with the two-look adjustment gives $p = 0.062$, which does not clear 0.05. The full-$n = 32$ registered result survives the two-look check ($p = 0.021$; Appendix B). As a linear point predictor, however, divergence adds little: leave-one-seed-out squared prediction error is only 5% lower than that of the training-seed mean predictor. This is cross-validation within the same cohort, not prospective validation on another task or cohort. At the interim look ($n = 16$) the same correlation was −0.24 and not significant; the pre-registered n was chosen for $|\rho| \approx 0.5$.

What the predictor result does and does not show.

- *What we found.* Agents whose policy has drifted further from their clean fork by the time the restriction is removed tend to recover more slowly. The association is pre-registered and survives correction across four candidates. Its direction remains without the extreme collapsed agent, although the combined exclusion-plus-two-look sensitivity is weaker. A cheap, reliable (split-half 0.95) removal-time measurement therefore carries rank information about recovery cost.

- *What we do not claim.* (i) Causality. In the exploratory drift-suppression arm (Section 5, $n = 4$, a different deficit window), a KL anchor to the onset policy during the restriction did not detectably speed recovery. That arm limits a related form of divergence, not divergence from the clean fork. Divergence may therefore mark slow recovery rather than cause it; a causal test, for example anchoring toward a clean fork, is left to future work. (ii) Useful point prediction: the leave-one-out gain is +5%. (iii) Failure of the other three measurements: their intervals still include moderate negative correlations. Detecting $\rho = -0.4$ with 80% power under Holm correction needs about 59 agents.

In the pilot, which varied duration, no measurement predicted recovery beyond duration. The supported association is within a single duration.

Descriptively, early–late recovery-cost correlations are near zero in the pilot ($\rho = -0.01$, $n = 4$) and Phase 8 (-0.05, $n = 8$). These small samples do not establish independence or rule out a useful association. The outcome-reliability analysis is post-hoc. Its pilot estimate (Spearman–Brown 0.30, from 4-seed rankings at short durations) was contradicted by the Phase 8 seeds (0.95 at $d = 300$), and we withdrew a "predictor ceiling" interpretation from an earlier draft.

9 Does the localization generalize? (pre-registered)

Evidence status. The completed generalization tests have registered contrasts and a fixed criterion, generally with 8 seeds. They are new settings or new contrasts on reused seed IDs, not independent new-seed replications. The Phase 12 onsets were selected from separate control-only pilot runs; seeds 0–3 of those pilots also enter the tests. CartPole acute checkpoint values were visible during progress checks (Appendix B); any incomplete test is descriptive only. The gap–damage relation was first noticed **post-hoc** and then tested out of sample in Phase 12 under a pre-registered prediction.

All results so far come from one game, one deficit family and one network size. Before running anything, we pre-registered three generalization tests of the PPO 2^3 surgery, each with 8 seeds and the same analysis and fixed criterion. The localization criterion is met if the *same-update recovery gap* has an interval below 0, the actor main effect has an interval below 0, and the actor has the largest point-estimated Shapley share. This gate concerns the gap that the surgery decomposes, not damage beyond lost time. If that gap is not detectably negative, no carrier ranking is confirmed ("no gap to localize"). Such a result is unresolved, not evidence of equivalence; it is not counted among the positive localization tests. The three tests were:

- a second game (Asterix-MinAtar, no left/right, onset at the meaning-matched update 390 located from a long control);
- a different deficit family (reward sign flip with probability 1 during the late window, SpaceInvaders);
- a wider network (2×256 actor and critic, movement restriction, SpaceInvaders).

Second game. On Asterix the late movement restriction again caused unlearning during the window, but only a little: $-1.0\,[-1.6, -0.3]$, about 18% of the clean agent's competence. It was followed by complete recovery. The restricted agent ends slightly *ahead* of a paused control (delay-adjusted $+0.5\,[+0.0, +1.0]$), and there is no persistent damage to localize (total $-0.1\,[-0.5, +0.4]$). The short-lived deficit sits in the actor ($-1.0\,[-1.3, -0.6]$ at +10 updates) and has gone by +50.

Different deficit. Flipping the reward sign is a far harsher deficit. The agent learns to avoid reward, and its unmasked return drops to exactly 0 at removal. Unlike the action restriction, it causes damage beyond lost time within the 150-update budget (delay-adjusted AUC $-18.6\,[-27.2, -9.9]$). The carrier is the same: the actor's Shapley value equals the total damage (-32.4 of -32.3), neither critic nor optimizer state contributes detectably, and a clean-actor transplant restores the damaged agent to the clean agent's level ($-0.4\,[-3.4, +2.5]$ vs. clean).

Table 12: Registered PPO carrier-generalization tests (paired 95% t-intervals). The same-update gap includes lost training time; delay-adjusted AUC is relative to the pause baseline. Actor share is a ratio of Shapley and total means, not a fraction of encoded information, and can exceed 100% when other components help. No share is interpreted when the mean gap is not detected. The criterion is defined in the text. Prior sources are the Phase 8 surgery analyses and `p9_asterix_rflip_surgery_n8-analysis.md`; Phase 12 sources: `p12_panel-analysis.md`. The Asterix flip was completed after an unblinded $n = 4$ look.

setting	n	same-update AUC gap	delay-adj. AUC	actor share	criterion
SpaceInvaders, movement, 2×128	12	$-15.1\,[-17.2, -13.0]^*$	–	90%	replicates
Asterix, movement, 2×128	8	$-0.1\,[-0.5, +0.4]$	$+0.5\,[+0.0, +1.0]^*$	n/a	no gap to localize
SpaceInvaders, reward flip, 2×128	8	$-32.3\,[-40.7, -23.9]^*$	$-18.6\,[-27.2, -9.9]^*$	100%	replicates
Asterix, reward flip, 2×128 (Phase 9)	8	$-2.3\,[-2.9, -1.7]^*$	$-1.7\,[-2.1, -1.3]^*$	94%	replicates
SpaceInvaders, movement, 2×256	8	$-19.7\,[-23.7, -15.6]^*$	$-5.1\,[-8.3, -1.8]^*$	82%	replicates
Phase 12 panel (pre-registered; critic and optimizer TOST at ±10% of the clean continuation's AUC)					
CartPole, reward flip (G1)	8	$-223\,[-347, -99]^*$	$-197\,[-323, -70]^*$	95%	met
Acrobot, reward flip (G2)	8	$-109\,[-237, +19]$	$-54\,[-147, +39]$	–	no gap detected
SpaceInvaders, actuator noise (G3)	8	$-9.5\,[-12.7, -6.3]^*$	$+4.2\,[+0.9, +7.5]^*$	115%	met
Freeway, reward flip (G4)	4	$-38\,[-40, -36]^*$	$-34\,[-35, -32]^*$	79%	descriptive only
SpaceInvaders, observation noise (G5)	8	$-10.0\,[-13.4, -6.5]^*$	$+3.7\,[+0.5, +7.0]^*$	107%	met
SpaceInvaders, reward extinction (G6)	8	$-9.6\,[-13.4, -5.8]^*$	$+4.1\,[+0.9, +7.4]^*$	104%	met

Wider network. Doubling the width of both networks (2×256) leaves the picture unchanged. The late movement restriction causes unlearning ($-28.7\,[-36.4, -21.0]$) and a recovery deficit of -19.7 AUC. The actor carries 82% of it (Shapley $-16.1\,[-18.0, -14.1]$), the critic again adds a small, early contribution ($-2.9\,[-5.4, -0.4]$, 15%; -10.1 at $+10$ updates, $+1.0$ at $+150$), and Adam state is equivalent to zero within ± 3 (TOST $p = 0.013$). A clean-actor transplant rescues $+15.7\,[+11.7, +19.7]$.

Phase 12 panel: more environments, more deficits, a second learner. In response to a fourth review round we pre-registered seven more tests before any of their data existed (adeb6b0; addenda 92fd75b and bf9a946). Each PPO test was planned at 8 seeds with the same surgery and criterion; five reached 8, while Freeway stopped at 4 complete seeds and is descriptive only. Onsets come from the same meaning-matched rule, measured from the untrained level so that it also applies to tasks whose returns are not near zero at initialization.

- *Environments*: two classic-control tasks without pixels (CartPole, Acrobot) and a third MinAtar game (Freeway), each under the reward flip.
- *Deficit families*, all on SpaceInvaders: *actuator noise* (with probability 0.5 the executed action is replaced by a uniformly random one, while the agent stores its own choice), *observation noise* ($\sigma = 1$) and *reward extinction* (reward set to 0).
- *Learner*: PQN-R under the reward flip, which writes wrong rewards into the replay buffer (G7).

The PPO rows are at the bottom of Table 12.

CartPole (G1). The flipped agent returns almost the minimum episode score: at removal it has lost 98% of its competence. Harm persists beyond lost time (delay-adjusted AUC -197^*). The actor accounts for 95% of the *same-update* recovery gap (-223; 8/8 seeds), and critic and optimizer state are each equivalent within the scale-dependent margin. This fork-reference surgery does not separately attribute the pause-relative estimand.

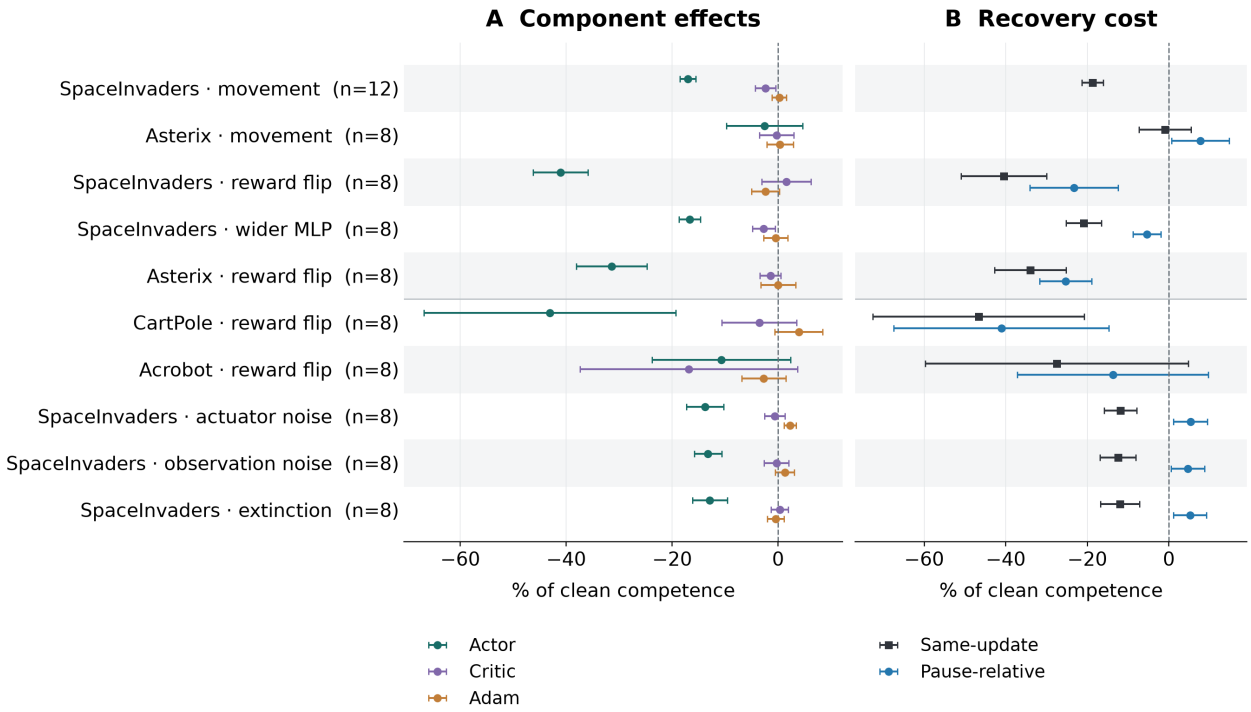


Figure 6: Expanded PPO generalization. **A**: operational actor/critic/Adam main effects. **B**: the same-update recovery gap and pause-relative recovery cost. Points are paired means with 95% t-intervals, scaled by each setting's mean clean AUC (Acrobot: clean competence above −500). This is visualization, not pooled inference; normalization does not equalize task difficulty. Missing pause-relative points are unavailable for the main $n = 12$ surgery. The line separates prior settings from Phase 12. Only complete cohorts are plotted; descriptive Freeway is shown separately in Table 12. Raw component contrasts and Holm tests are in Appendix Table 19.

Acrobot (G2). At the meaning-matched onset the agent is still weak (−346, against −500 for the untrained policy). The change during the deficit is +18 [−51, +87]: unlearning is not detected, but the interval is wide. A plausible reading is delayed improvement rather than catastrophic loss of prior skill; we did not separately measure that mechanism. Total damage is −109 [−237, +19], so there is no damage to localize. The intervals are very wide and neither the critic nor the optimizer is equivalent to zero, so this null is weak evidence. The critic's point estimate is larger than the actor's, although both intervals include zero; actor dominance is not established for Acrobot.

Actuator noise (G3). Random actuator faults cause unlearning (−14.5*) and a same-update recovery deficit (−9.5*). The actor carries all of it (Shapley share 115%, 8/8 seeds); the damaged Adam state slightly *helps* (+1.8*). As under the movement restriction, there is no damage beyond lost time: the restricted agent ends ahead of the pause reference (delay-adjusted AUC +4.2*).

Observation noise (G5). Gaussian noise on the observations has the same signature as actuator noise. It causes unlearning (−11.6*) and a same-update deficit (−10.0*) that the actor carries (107%, 8/8 seeds), with critic and optimizer state equivalent to zero. There is no damage beyond lost time (delay-adjusted AUC +3.7*), consistent with our earlier observation-noise pilot (Section 4).

Reward extinction (G6). Setting the training reward to zero also causes unlearning (−14.8 [−24.5, −5.0]) and a same-update gap (−9.6 [−13.4, −5.8]). The actor is the largest carrier (104%, 8/8 seeds), with no detected critic or Adam harm. The pause-relative AUC is positive (+4.1 [+0.9, +7.4]), so the observed cost is not persistent harm beyond lost time. This extends the actor-localization pattern to a deficit with no explicit incentive to avoid the task's true reward.

Table 13: PQN-R reward-flip surgery ($n = 8$). Primary: +10..650 updates; registered secondary: $+20, +40, \ldots, +300$, matched to PPO in environment steps. Paired 95% t-intervals; * denotes an unadjusted interval excluding zero. Source: `p12_g7_pqnr_rflip-analysis.json`.

contrast	primary AUC	matched-horizon AUC (secondary)
same-update gap	$-6.78\,[-9.42, -4.15]^*$	$-11.85\,[-15.11, -8.59]^*$
network-block main effect	$-6.22\,[-8.32, -4.13]^*$	$-9.71\,[-11.94, -7.48]^*$
buffer main effect	$-0.56\,[-1.73, +0.61]$	$-2.14\,[-4.00, -0.28]^*$
clean-buffer rescue	$+1.74\,[+0.19, +3.28]^*$	$+3.69\,[+1.05, +6.33]^*$
interaction	$-2.35\,[-4.51, -0.19]^*$	$-3.11\,[-6.53, +0.32]$

Freeway (G4, incomplete; descriptive only). The four complete seeds lose all competence at removal and show no recovery within the measured window. The same-update gap is $-38.0\,[-40.1, -35.9]^*$, and the pause-relative AUC is $-33.5\,[-35.3, -31.8]^*$. The actor has a 79% point-estimated share; the critic also has a substantial effect ($-7.3\,[-11.5, -3.1]^*$). This is evidence to report, not a registered confirmation: only four of the planned eight seeds completed all arms, and partial arms for seeds 4–5 are excluded from the factorial aggregate.

Algorithm axis: PQN-R under reward flip (G7, registered, $n = 8$). The sign-flipped rewards are stored in the 100k replay buffer during the same 409,600-step deficit as the PPO tests. At the prespecified 650-update recovery horizon, the same-update gap is $-6.8\,[-9.4, -4.2]$, and the joint network block (parameters *and* RAdam state) accounts for 92% of it. The buffer main effect is $-0.6\,[-1.7, +0.6]$ and meets the registered 10%-of-clean-AUC equivalence rule (± 4.93, TOST $p < 10^{-4}$). This coarse margin does not establish equivalence within one point and does not replace the matched-identity test of Section 7.

The registered secondary grid exactly matches PPO's 15 recovery checkpoints in environment steps: PQN-R $+20, +40, \ldots, +300$ equals PPO $+10, +20, \ldots, +150$. On this shorter grid the buffer effect is detected ($-2.1\,[-4.0, -0.3]$), with a share of 18%. Restriction-era transitions leave the 100k buffer by +25 updates. The result therefore supports network dominance while showing that a buffer contribution can be obscured by averaging over a longer recovery horizon. A clean buffer helps an already damaged network, and the primary interaction is negative; we do not interpret the buffer as uninvolved. This whole-buffer transplant does not isolate reward-label identity from the changed state/action distribution, and no parameter-versus-moment split was run under this new deficit.

Historical pause benchmark (post-hoc). The standardized-context DD continuation compared with C shifted by 100 updates gives full-horizon AUC $-2.48\,[-4.98, +0.03]$, matched-horizon AUC $-4.81\,[-7.78, -1.85]$, and last-five-checkpoint difference $+0.52\,[-3.40, +4.43]$. These are finite-horizon benchmarks, not a new network/buffer attribution of the pause-relative estimand. The full-horizon and late comparisons are inconclusive; shorter-window harm is not proof of irreversible failure.

Why Asterix showed no lasting damage under the movement restriction. Asterix is counterevidence to any claim that late deficits *always* cause lasting damage, and we make no such claim. Two features of that setting predicted a null under the lost-competence account of Section 4.

- *Little competence was lost.* Removing left and right in Asterix still leaves up and down movement, so the restricted policy kept most of its competence (gap at removal 18%, against 63–100% in the SpaceInvaders settings).
- *There was no progress to lose.* The clean agent made no progress during the window (5.36 to 5.41), so the restricted agent had no lost progress to make up. In SpaceInvaders the clean agent improved by 28–45% in the same window.

To test whether the null was specific to the game or to the deficit, we pre-registered the same surgery on Asterix with the harsher reward-flip deficit (Phase 9, 8 seeds; completed after an interim look at 4 seeds,

Table 14: Post-hoc (descriptive): competence lost at removal vs. damage across the PPO surgery settings. Relative quantities are divided by the clean agent's competence at removal $C(w_1)$. Clean progress: $C(w_1) - C(w_0)$ during the window. * SpaceInvaders 2×128 has no evaluation during the window for its 12 surgery seeds; $C(w_0)$ and $C(w_1)$ are from the Phase 2 seeds 0–3. Source: `p9_gap_vs_damage-posthoc.md`.

setting	n	clean progress	gap at removal	relative gap	same-update gap	relative total	delay-adj. AUC
Asterix, movement	8	+1%	-1.0^*	18%	$-0.1\,[-0.5, +0.4]$	−2%	$+0.5\,[+0.0, +1.0]^*$
SpaceInvaders, movement, $2 \times 128^*$	12	+29%	-37.0^*	63%	-15.1^*	−26%	–
Asterix, reward flip	8	+1%	-3.7^*	68%	-2.3^*	−43%	$-1.7\,[-2.1, -1.3]^*$
SpaceInvaders, movement, 2×256	8	+45%	-49.9^*	73%	-19.7^*	−29%	$-5.1\,[-8.3, -1.8]^*$
SpaceInvaders, reward flip	8	+28%	-58.2^*	100%	-32.3^*	−55%	$-18.6\,[-27.2, -9.9]^*$

Table 15: Phase 12: out-of-sample test of the gap–damage relation (pre-registered prediction, 92fd75b). Relative gap and relative total are divided by the clean agent's competence at removal (Acrobot: by its competence above the untrained level, because its returns are negative).

setting	n	relative gap	same-update gap	relative total	delay-adj. AUC	prediction; outcome
CartPole, reward flip (G1)	8	98%	$-223\,[-347, -99]^*$	−49%	$-197\,[-323, -70]^*$	gap; gap (hit)
Acrobot, reward flip (G2)	8	44%	$-109\,[-237, +19]$	−35%	$-54\,[-147, +39]$	no prediction; no gap
SpaceInvaders, actuator noise (G3)	8	47%	$-9.5\,[-12.7, -6.3]^*$	−16%	$+4.2\,[+0.9, +7.5]^*$	no prediction; gap
Freeway, reward flip (G4)	4	100%	$-38\,[-40, -36]^*$	−123%	$-34\,[-35, -32]^*$	gap (partial); descriptive only
SpaceInvaders, observation noise (G5)	8	42%	$-10.0\,[-13.4, -6.5]^*$	−17%	$+3.7\,[+0.5, +7.0]^*$	no prediction; gap
SpaceInvaders, reward extinction (G6)	8	47%	$-9.6\,[-13.4, -5.8]^*$	−16%	$+4.1\,[+0.9, +7.4]^*$	no prediction; gap

which gave the same result). The flip removed 68% of competence by removal ($-3.7\,[-4.6, -2.8]$), while the clean agent again made no progress. This time there was lasting damage beyond lost time: delay-adjusted AUC $-1.7\,[-2.1, -1.3]$, still $-1.2\,[-1.7, -0.6]$ at the end of the 200-update budget. The total was $-2.3\,[-2.9, -1.7]$, and the localization criterion was met. The actor's Shapley share is 94% (8/8 seeds), critic and optimizer state are each equivalent to zero within ± 3 (TOST $p < 10^{-4}$), and a clean actor restores the clean agent's level ($+0.0\,[-0.4, +0.5]$ vs. clean). The Asterix null was therefore a property of the mild deficit, not of the game.

Post-hoc, across settings (Table 14; descriptive, five settings): total damage grows with the fraction of competence lost at removal. It is −2% of the clean agent's competence at a gap of 18%, −26 to −43% at 63–73%, and −55% at 100%. Damage beyond lost time appears in the settings with the largest gaps. The pattern is consistent with a simple account: lasting damage requires that the deficit destroys more competence than can be relearned within the budget, plus whatever progress the clean agent makes meanwhile. Whenever damage of either kind exists, the actor carries most of it.

Out-of-sample test (pre-registered, Phase 12). After CartPole acute values were visible, but before its final recovery analysis and before the other panel outcomes were inspected, we committed a prediction from this post-hoc relation (92fd75b). CartPole is therefore displayed separately, not counted as an untouched validation. Settings losing less than 25% of competence at removal were predicted to show no detectable same-update recovery gap, and settings losing 60% or more to show a gap; for 25–60% we made no prediction. This does not predict harm beyond lost training time. Table 15 gives the outcome. Only CartPole among the completed new settings falls in a prediction band, and its outcome is consistent with the prediction; its acute values had already been seen. All four other completed settings fall in the intermediate band, for which no prediction was issued. The incomplete Freeway sample is consistent descriptively but is not counted as a confirmatory validation. Thus there are no completed, untouched new-setting predictions with which to establish this relation; it remains a post-hoc hypothesis, not a general severity law.

Before Phase 12, the four settings with a detected same-update gap all had the actor as largest point-estimated carrier (82–100%). In the two SpaceInvaders movement settings the critic added a small early share (12%, 15%); in the two reward-flip settings it was not detected. Across all ten completed PPO settings, eight have a detected same-update gap and all eight meet the registered actor criterion (82–115%). The two unresolved settings are Asterix movement and Acrobot reward flip. Freeway, at $n = 4$ of the planned 8, is excluded from this confirmatory tally. The additional panel-wide actor Holm sensitivity and covariance-aware equivalence checks leave the completed-setting calls unchanged (post-hoc; `p12_audit-posthoc.md`).

10 Discussion

Scope of the conclusions. Every conclusion in this paper is about the following tested space. Phrases such as "in our settings" refer to it, and we do not extrapolate beyond it.

- *Learners.* PPO (all carrier-generalization tests), PQN and PQN-R (timing; network versus buffer under the movement restriction and, in Phase 12, under the reward flip). All are small MLPs on flattened observations (2×128; one 2×256 test). There are no convolutional, recurrent or large-scale agents.
- *Environments.* Four tasks have complete carrier tests: SpaceInvaders-MinAtar (main; its gymnax dynamics differ from MinAtar), Asterix-MinAtar, CartPole-v1 and Acrobot-v1. Freeway-MinAtar has a descriptive four-seed subset of a planned eight-seed test. There is no continuous control, no full-scale Atari, and no hard-exploration or procedurally generated task.
- *Deficits.* Action restriction (two kinds), reward sign flip, actuator noise, observation noise and reward extinction. Each was applied once, for 50 updates (409,600 steps) at a meaning-matched onset, and in SpaceInvaders the restriction was extended to 300 updates. Untested: task switches, other dynamics shifts, data starvation, repeated or overlapping deficits.
- *Budgets.* Finite recovery horizons, typically 1.2M–3.7M environment steps (150–250 PPO updates, 650–900 PQN updates), on a 2-core CPU. We do not test irreversible damage or unlimited recovery. Budgets and checkpoint grids are recorded per study.

The main movement restriction: savings rather than slower same-task relearning. For PPO under the SpaceInvaders movement restriction, the fresh-seed estimand test confirms faster relearning from equal competence. The matching timing patterns for PQN and PQN-R remain exploratory ($n = 4$). For this intervention, the evidence is consistent with the following reading:

- early deficits cost time;
- late deficits cause unlearning that is relearned, and from equal competence restricted agents relearn faster than clean agents first learned (a savings effect confirmed on fresh seeds, Table 5);
- in the main PPO probe, the prior policy remains decodable on a fixed state bank; in the four-seed reset test the damaged actor outperforms a fresh one;
- what persists is the progress the clean agent made in the meantime, stored in the function approximator.

This argues against a *persistent relearning deficit for the main movement restriction within its recovery budget*. Reward flips in SpaceInvaders, Asterix and CartPole do produce pause-relative harm; we did not run a competence-matched rate test for all those settings. It does not show that plasticity in the broader sense is preserved. We did not measure learning on new tasks or under further distribution shifts, nor standard plasticity diagnostics (dormant units, feature rank, gradient statistics). A deficit could leave the agent able to relearn its old competence quickly while impairing what it can learn next. Both the savings effect and the reset result are indirect evidence. In PPO the main carrier is the actor's parameters (Shapley shares are summarized over settings with a detected same-update gap in Section 9). Under the SpaceInvaders movement restriction the critic is a minor carrier: significant in the pooled $n = 12$ (12%) and with the wider network (15%), concentrated in early recovery; the fresh-only critic interval at $n = 8$ includes zero. The main-study Adam contribution is equivalent to zero within ± 3. In the actuator-noise test Adam state instead gives a small positive contribution ($+1.8$), so "equivalent within a practical margin" must not be confused with "uninvolved". These results complement optimizer-state accounts in other settings (Dohare et al., 2023; Xu, 2026). In PQN-R under the movement restriction, the parameter-versus-moment split locates most of the network effect in the parameters; the primary optimizer-only rescue is inconclusive, with a small secondary effect and a substantial short-lived interaction. A large buffer extends the period during which restricted data shape the network. Its effect outlasts the data, which in this configuration suggests that it mediates the damage rather than storing it.

Relation to the primacy bias and resets. Nikishin et al. (2022) found that resetting parts of an agent while keeping its buffer cures early overfitting. In our setting resets are harmful: an actor reset costs -27.2. The difference is informative. Primacy-bias damage is a network fitted to stale early data whose capacity to fit new data is impaired. Under our movement restriction, the PPO reset experiment instead concerns an agent that can quickly regain prior competence. A reset discards both the damage and the retained competence, and the second matters more.

Practical implications. These are suggestions from the tested protocols, not general prescriptions for RL agents:

(i) in the four-seed PPO SpaceInvaders reset experiments, a fresh actor was a worse recovery starting point; this argues against blindly resetting in that setting, not against resets on new tasks or in other agents;

(ii) protecting the policy during the deficit (freezing, KL anchoring) preserves evaluation performance during the deficit but did not detectably speed recovery (gains above $+5.7$ and $+1.7$ AUC are excluded, $n = 4$);

(iii) in replay agents, refreshing the buffer with fresh, competent data helps modestly, and keeping the replacement resident longer matters about as much as which data is removed;

(iv) a clean policy from a matched parallel run is an effective diagnostic transplant in these PPO tests. Briefly training on its data with importance weighting also helps; without weighting the same data are harmful. This oracle clean fork is not generally available in deployment.

Limitations.

- **Scope.** The tested space is listed at the start of this section. Most mechanistic detail (timing, PQN-R, the replay ladder, the point of no return) comes from SpaceInvaders-MinAtar in gymnax, whose dynamics differ from MinAtar. The carrier tests cover four complete tasks, one incomplete task and five deficit families, mostly with PPO and small MLPs. Completed generalization tests have 8 seeds; Freeway has only 4 complete seeds. The optional independent-seed replications did not run, so the new generalization evidence is from registered new contrasts on reused IDs, not independent new-seed replication. Lasting damage is not universal: it was absent when the deficit removed little competence. Larger convolutional agents, high replay ratios (Schwarzer et al., 2023), continuous control, longer training and other deficits may behave differently.

- **Seeds.** The main PPO surgery has 12 seeds and completed generalization tests 8; incomplete Freeway has 4. Several secondary results (drift suppression, PQN early/late recovery, fire restriction) still rest on 4 seeds. Appendix Table 21 gives their minimum detectable effects, and we phrase them as bounds.

- **Point of no return and predictors.** Both pre-registered tests at $d = 300$ were inconclusive. Average persistent deficits of a few return points are not excluded, and the frequency of the post-hoc extreme-collapse phenotype (1 of 44) is estimated very imprecisely. The non-recovery diagnosis (policy collapse) is post-hoc and rests on a single agent. Its recovery was tracked only within a 250-update budget. The policy-divergence predictor is an association with small out-of-sample gain. The predictor test was completed after an unblinded interim look (disclosed in Appendix B), and the null results for the other three predictors are bounds.

- **Data versus update rule in PPO.** The ratio-neutral swap removes importance weighting, but it applies on-policy updates to off-policy actions. Importance weighting and clipping change what a behaviour swap does. No PPO arm cleanly isolates data content from the update rule, and we draw no conclusion about the effect of data content alone in PPO.

- **Interventions.** Two deficit families, removing actions and flipping the reward sign, actively push the policy away from its competence. Phase 12 added actuator noise, observation noise and reward extinction, without an explicit adversarial reward objective, but tested each in one setting only. Other deficits may produce different damage: data starvation, other non-stationary dynamics, task switches that cause catastrophic forgetting, or repeated deficits. The same applies to surgery: transplants create hybrid states that do not occur in natural training (Section 5). Component effects are defined relative to that procedure, and the critic's effect depends on its context.

- **Plasticity.** Our evidence for relearning preserved competence is indirect and task-specific: savings and the reset result. It is not a test of plasticity in general. Learning capacity on new tasks was not measured.

- **Mechanistic readings.** "Carrier" is an operational, interventional notion (Section 3). "The buffer acts through the network" and "the actor retains useful structure" are interpretations of intervention results in these setups. They are not direct measurements, and they should not be generalized beyond the tested agents, buffers and deficits.

- **Effect scales.** AUC is measured in raw evaluation return, and PPO and PQN-R recover over different horizons and reach different returns. Raw AUC effects are therefore not comparable across algorithms. Normalization in Appendix Table 23 aids interpretation but does not equalize horizons or task difficulty. Phase 12 G7 additionally reports a recovery grid exactly matched to PPO in environment steps; it does not constitute a complete algorithm-by-task factorial.

- **Incomplete confirmatory evidence.** Several conclusions remain descriptive or exploratory (Table 2): the PQN and PQN-R timing results and the drift and reset arms ($n = 4$), and every analysis labelled post-hoc. Only the results marked confirmatory met their pre-registered decision rules.

- **Counterfactual baseline.** Numerical "damage" depends on the counterfactual. We fix one estimand per question (Section 3.4) and test both primaries confirmatorily only for PPO's late movement restriction. For the other learners the comparison is post-hoc and rests on 4 seeds. The pause baseline assumes that a restricted update is worth nothing, whereas some learning does happen under the restriction (early windows, Table 3). It can therefore favour the restricted agent slightly. The fork-reference surgery decomposes the same-update deficit. The onset-reference surgery decomposes a state-level pause counterfactual under the standardized future context, complementary to the historical delay-adjusted primary, and it was run for one deficit with damage beyond lost time ($n = 8$) and descriptively for the movement restriction ($n = 4$).

- **Replay identity is bounded, not zero.** The rung-4 equivalence result clears its ± 1 margin only narrowly (TOST $p = 0.0495$), with a positive point estimate ($+0.46$). We claim only that identity effects of ≥ 1 AUC point are excluded. Smaller effects remain possible, and the results do not establish that replay-data identity has no effect.

- **Exploratory early phases.** The PPO and PQN phases before the purge study preceded our pre-registration practice. Their central PPO claim has since been replicated under pre-registration.

11 Conclusion

We summarize by evidence strength (Table 1); all statements concern the tested space listed under "Scope of the conclusions" in the Discussion (small MLP agents, four complete tasks and one incomplete task, CPU-scale budgets).

- *Confirmed.* In PPO, the recovery deficit after a late deficit is carried mainly by the actor's parameters, in the operational sense of state surgery. The critic adds a small, transient share and Adam state adds nothing detectable. The onset-reference test also localizes pause-relative harm under the SpaceInvaders reward flip. Across the eight completed PPO settings with a detected same-update

gap, the actor has the largest point-estimated share (82–115%); this does not establish actor dominance in every task. Acrobot is unresolved, and the Freeway result is descriptive at $n = 4$. A registered reward-flip test in PQN-R ($n = 8$) also supports the joint network block as the largest carrier of the same-update gap (92%); it does not separately test parameters versus moments in that new setting. Buffer effects depend on recovery horizon. Under the main movement restriction, fresh-seed PPO agents relearn faster than clean agents first learned. Policy divergence from the clean fork at removal is associated with slower recovery.

- *Bounded.* In PPO under the movement restriction, damage beyond lost time larger than 3 AUC points is excluded. No average point of no return was detected after 2.46M restricted steps, and average deficits worse than about 5 points are excluded. In the matched PQN-R movement-restriction identity test, a 1-point effect is narrowly excluded by the prespecified TOST; smaller effects are not.

- *Exploratory.* Early restrictions cost only time, across PPO, PQN and PQN-R (4 seeds each). In PQN-R, the Q-network rather than the buffer carries most of the damage under the movement restriction.

- *Post-hoc.* Whether lasting damage occurs tracks how much competence the deficit destroys (tested out of sample in Phase 12: no completed untouched setting fell in a prediction band; the relation remains provisional). One of 44 long-restricted agents collapsed and did not recover within the budget; how common such failures are is unknown (95% interval 0.06–12%).

These results locate damage in components; they do not reveal how it is encoded, and they do not establish that plasticity is preserved beyond relearning the same task. We hope the protocol (matched forks, unmasked common-random-number evaluation, delay-adjusted recovery and factorial state surgery) is useful for locating the carriers of other training pathologies in RL.

Broader impact statement

This is basic research on training dynamics in small simulated environments. We do not foresee direct negative societal impacts. Understanding how agents recover from temporary constraints may inform the safe deployment of RL systems in which actions are sometimes disabled.

Reproducibility statement

All code, configurations, run manifests (with commit hashes and package versions), per-seed results, analysis scripts and a chronological experiment log with every pre-registration are available at `[anonymized repository]`. Pinned versions: jax 0.11.2, flax 0.12.10, optax 0.2.8, gymnax 0.0.9, numpy 2.5.4. Runs are deterministic on CPU given the seed. Each experiment fits on a 2-core CPU without a GPU; for example, the PPO recovery test took 1,883 s and the point-of-no-return pilot 2,683 s. Appendix B maps every table to its source files.

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A Hyperparameters

Table 16: Agent hyperparameters (SpaceInvaders-MinAtar, flattened $10 \times 10 \times 6$ observations).

	PPO	PQN / PQN-R
network	separate actor/critic MLP 2×128, ReLU	Q-MLP 2×128, LayerNorm, ReLU
initialization	orthogonal ($\sqrt{2}$; heads 0.01 / 1.0)	Flax default
environments $\times$ steps	64×128 (8,192 per update)	128×32 (4,096 per update)
optimization	4 epochs $\times$ 8 minibatches	2 epochs $\times$ 32 minibatches
optimizer, LR	Adam, $5 \cdot 10^{-4}$ constant	RAdam, $5 \cdot 10^{-4}$ constant
gradient clip (global norm)	0.5	10
discount, trace	$\gamma = 0.99$, GAE $\lambda = 0.95$	$\gamma = 0.99$, Q(λ=0.65) (PQN)
loss terms	clip 0.2, entropy 0.01, value 0.5	$\frac{1}{2}$ squared TD error
exploration	stochastic policy	ϵ: $1 \to 0.05$ over 1M steps
replay (PQN-R)	–	FIFO 100k or 1M, batch 128, 1-step, replay ratio 2, no target net
early / late window	updates $[0, 50)$ / $[100, 150)$	updates $[0, 100)$ / $[250, 350)$
window length	409,600 env steps	409,600 env steps
evaluation	every 10 updates, 64 episodes, stochastic	every 10 updates, 64 episodes, $\epsilon = 0.05$

B Provenance, pre-registration timeline and deviations

Every number in the main text comes verbatim from an analysis file in `results/summary/` of the repository, after the sign conversion stated in Section 3. Each analysis reads only committed run summaries, and each run manifest records its commit hash, whether the working tree was clean, configuration and package versions.

Deviations and corrections.

(1) Early PPO analyses used percentile bootstrap intervals over 4 seeds. These were replaced by paired t-intervals. This reversed one conclusion: the PPO delay-adjusted residual at +250 updates changed from significant to $+2.0\,[-0.1, +4.1]$.

(2) The purge study's E arm (purge on the clean network) was added by us as a secondary arm before data collection. The B−E contrast was computed post-hoc.

(3) In Phase 6c, the factorial ANOVA was requested after the results and is labelled post-hoc. Only its interaction term had been pre-registered, as a descriptive secondary.

(4) In Phase 7b, secondary item 1 (the predictor metrics at $+10/+20/+30$ updates of recovery) was not run. Its outcome variables have the same low reliability as the primary ones, so it could not change the verdict. The run is deterministic and can be reproduced.

(5) The outcome-reliability analysis of Section 8 is post-hoc. Its pilot estimate (0.30) was contradicted by the Phase 8 seeds (0.95 at $d = 300$), and we withdrew the "predictor ceiling" interpretation from an earlier draft.

(6) Phase 8 (reviewer response). All runs, contrasts, margins and decision rules were committed before data (8c62fe0). The analysis scripts were written during the runs and implement only those contrasts; the baseline comparison, the rung-4 sensitivity analysis and the minimum-detectable-effect table are post-hoc. Twice the run queue stopped because the paper source or an untracked script made the working tree dirty, which the provenance guard refuses. No run was affected, and the queue resumed after a commit. The reproduction check against the Phase 2 reference is valid only for seeds 0–3, since the reference contains no other seeds. Seeds 0–3 matched exactly, and the Phase 8 DDD cells reproduced the earlier arms bit for bit.

(7) Phase 9 (second reviewer response). Estimands, runs, margins, tests and decision rules were committed before data (63a8f1d), together with analysis scripts validated on earlier data only. E1 ran

Table 17: Sources of the reported results.

result	analysis file (`results/summary/`)	design status
Table 3	`p2_recovery_all-analysis.md`	exploratory
Table 4	`p5_recovery_move-analysis.md`	exploratory; metric fixed before analysis
Table 6	`20261011-p3-surgery-move-late-analysis.md`, `20261011-p4-phase4-move-late-...-analysis.md`, `20261011-p4-phase4-fire-late-...-analysis.md`	exploratory
Table 8	`p6_surgery_pqnr_n12-analysis.md`, `p6_surgery_pqnr1m_n8-analysis.md`, `*-phases.md`	assignment rule pre-specified
Table 9, rung 1	`p6_purge-analysis.md`, `p6_purge-posthoc.md`	pre-registered (864697e)
rung 2	`p6b_targeted-analysis.md`	pre-registered; runs at 250cec4
rung 3	`p6c_2x2-analysis.md`, `p6c_2x2-factorial-posthoc.md`	pre-registered; runs at 2559f2e; ANOVA post-hoc
rung 4	`p6d_idswap-analysis.md`	pre-registered (a873dc0), n and SESOI fixed
Table 10	`p7_ponr_pilot-analysis.md`	gate pre-registered (ba1802f)
predictors	`p7b_predictors-analysis.md`	pre-specified (aabd1ae), script 39c3049
Table 6 ($n = 12$), data arms	`p8_ppo_surgery_n12-analysis.md`	pre-registered (8c62fe0); seeds 0–3 exploratory
PQN-R parameters vs. optimizer	`p8_pqnr_split-analysis.md`	pre-registered (8c62fe0)
confirmatory $d = 300$	`p8_ponr_confirm-analysis.md`	pre-registered (8c62fe0)
Table 12	`p8_asterix_surgery-analysis.md`, `p8_rflip_surgery-analysis.md`, `p8_h256_surgery-analysis.md`	pre-registered (8c62fe0)
Table 5	`p9_estimands-analysis.md`	pre-registered (63a8f1d)
replication $d = 300$, Table 11	`p9_ponr_predictors_n32-analysis.md` (interim: `p9_ponr_predictors-analysis.md`)	pre-registered (63a8f1d); completed in Phase 11
Asterix reward flip	`p9_asterix_rflip_surgery_n8-analysis.md` (interim: `p9_asterix_rflip_surgery-analysis.md`)	pre-registered (63a8f1d); completed in Phase 11
Phase 12 panel, Tables 12 and 15	`p12_panel-analysis.md`, `p12_<test>-analysis.md`, `p12_tables.tex`	pre-registered (adeb6b0; addenda 92fd75b, bf9a946)
Phase 12 long controls (onsets)	`20261014-p2-p12-longctl-*.json`	onset rule fixed in Phase 5, generalized in adeb6b0
Phase 12 algorithm axis, Table 13	`p12_g7_pqnr_rflip-analysis.md/.json`	primary bf9a946; matched horizon 3ddffe2; pause benchmark post-hoc
interim/outlier combined sensitivity	`p12_interim_predictors-posthoc.md/.json`	post-hoc; full-sample primary unchanged
Table 14	`p9_gap_vs_damage-posthoc.md`	post-hoc
Table 7	`p10_onsetref_rflip-analysis.md`, `p10_onsetref_move-analysis.md`, `p10_onsetref_rflip-checks.md`	pre-registered (f618d6c)
Tables 24, 23	`p10_interactions_*-posthoc.md`	post-hoc
Tables 21, 22; rung-4 sensitivity	`p8_power-posthoc.md`, `p8_alt_baselines-posthoc.md`, `p8_6d_sensitivity-posthoc.md`	post-hoc

as planned. Two sandbox restarts cost about a third of the 6 h budget. With the authors' decision, recorded before any E3 outcome was analysed, the Asterix reward-flip test was cut to 4 seeds (completed to 8 in a later session, after the 4-seed result had been seen) and the point-of-no-return replication stopped at 16 of 32 seeds. An earlier draft reported those 16 seeds descriptively, as its pre-registered fallback required. In a later session (Phase 11; queue commit ac63459, which logged the reporting rule before the runs) both tests were completed to their pre-registered n with unchanged code and configuration. Each therefore had one unblinded interim look. We did not adjust for it, and we report the interim estimates alongside the final ones. For the replication the interim and final estimates differ: at $n = 16$, L_{300} was $+6.0\,[+1.9, +10.2]$ and no predictor was supported. The decision rules were applied only at $n = 32$. The diagnosis of the non-recovering seed 330 and the exclusion sensitivity analysis are post-hoc. All deviations are logged with timestamps (commits 4a02dd2 and later).

(8) Phase 12 (fourth reviewer response, generalization). The tests, onset rule, margins and decision rules were committed before any deficit run (adeb6b0). Two addenda followed, each before any data in its target test had been formally analysed, but after CartPole acute values had been seen during progress checks: an out-of-sample test of the gap–damage relation (92fd75b) and the PQN-R test G7 (bf9a946). Four deviations occurred, all logged with timestamps. (i) The first queue version gave both seed chunks of a test the same tag. It was stopped one minute into CartPole seeds 4–7; that partial run was deleted unanalysed and the chunk was rerun under per-chunk tags. (ii) Progress checks exposed CartPole's clean *and damaged* returns at removal, as well as clean-arm recovery checkpoints, before the final $n = 8$ primary analysis. The panel design and decision rules were already fixed. However, the gap–damage addendum came after these acute values were visible, so CartPole is reported separately rather than counted as an untouched validation of that addendum. An audit correction was committed (79c9c85). (iii) G7's chunk start cutoff was moved from 22:55 to 23:20 before any G7 data existed. (iv) A smoke test of the new PQN-R reward-flip path timed out during compilation and was deleted. (v) The first Freeway chunk (4 seeds) ran out of memory on the 2 GB machine and was killed. Freeway was rescheduled in 2-seed chunks (same seeds, same configuration), and its partial outputs were deleted unanalysed. (vi) While that chunk was being stopped, the queue's clean-tree guard rejected every remaining chunk, because a tracked file was modified at that moment. No runs were produced. The tests were requeued in the order G5, G6, G7, G4, so that the memory-heavy Freeway runs came last. A later sandbox reset terminated the queue during Freeway seeds 4–5. Only seeds 0–3 have all eight factorial arms; their analysis is descriptive, and incomplete arms for seeds 4–5 are retained in the data but not pooled into a factorial estimate. Seeds 6–7 and the optional independent-seed replications (600–607 for CartPole; 700–707 for actuator noise) never ran. No missing outcome was imputed or treated as a registered success. No formal interim primary-AUC analysis was performed: each test's final analysis was run only after all its chunks finished. This is distinct from the unblinded acute-outcome progress checks in (ii).

(9) Two conclusions of the earlier draft changed with more seeds or better controls: the PPO critic is not uninvolved (a small transient effect at $n = 12$), and the PPO data-swap benefit is not a data-content effect (it disappears without importance weighting).

Interim looks: what they could and could not have influenced. Two pre-registered experiments were completed after an unblinded interim look (Table 18). For both, the following were fixed in the Phase 9 pre-registration (63a8f1d) before any of their data existed: seeds (Asterix 0–7; replication 300–331), code, configuration, estimands, margins, tests and decision rules. None changed after the looks. The decision to complete them was taken at the start of a later session, on budget grounds, as the default continuation of the pre-registered plan. Before any new run, it was logged together with the reporting rule ("report the pre-registered decision at the pre-registered n, and report the interim values alongside it", commit ac63459). The log records the decision as completion of the original seed plan, not a result-triggered stopping rule. Nevertheless, it was made with knowledge of the interim results, and we cannot claim a blinded continuation. The data-prompted diagnostic analyses in that study were the diagnosis of seed 330 and the exclusion sensitivity analysis. The review-requested two-look and combined sensitivity checks were added later and are also post-hoc. They were prompted by the final data, are labelled post-hoc and do not enter any decision. An unblinded look can still inflate nominal error rates. As a conservative check we therefore apply a Bonferroni correction over the two looks (doubling each p). The final primary test decisions survive this check (Table 18); it is a sensitivity analysis, not a retroactive removal of all possible adaptive research or reporting choices.

C Additional results

Table 18: Interim looks and a conservative two-look adjustment. p-values are those of the pre-registered decision tests at the final n; "×2" doubles them (Bonferroni over looks).

experiment	interim (unblinded)	final (pre-registered n)	decision with $p \times 2$
Asterix reward-flip surgery	$n = 4$: total -2.4, actor share 93%	$n = 8$: total -2.3 $[-2.9, -1.7]$; actor Holm $p < 10^{-4}$; critic, Adam TOST $p < 10^{-4}$	unchanged (all $p < 2 \times 10^{-4}$)
replication, L_{300} (TOST ±5)	$n = 16$: $+6.0$ $[+1.9, +10.2]$	$n = 32$: $+2.3$ $[-4.6, +9.2]$, TOST $p = 0.21$: inconclusive	unchanged (inconclusive)
replication, predictors (Holm over 4)	$n = 16$: none supported (KL $\rho = -0.24$)	$n = 32$: KL $\rho = -0.48$, Holm $p = 0.010$; others $p \geq 0.27$	unchanged (KL $p = 0.021$)
combined sensitivity (post-hoc)	seed 330 excluded, $n = 31$	KL $\rho = -0.43$, Holm $p = 0.031$	$p \times 2 = 0.062$: not below 0.05

Table 19: Phase 12 PPO component effects (paired 95% t-intervals). * marks an unadjusted interval excluding zero; the actor column's Holm p corrects over the three effects within the setting. Critic/Adam equivalence uses the registered scale rule ±10% of |clean AUC|, not the older ±3 margin. Joint-scale and panel-wide multiplicity checks are post-hoc, reported in `p12_audit-posthoc.md`; they do not replace registered decisions.

setting	n	actor	critic	Adam	actor Holm p	margin
CartPole, reward flip (G1)	8	-206 $[-320, -92]$*	-17 $[-51, +17]$	$+19$ $[-3, +40]$	0.011	±47.9
Acrobot, reward flip (G2)	8	-43 $[-94, +9]$	-67 $[-148, +15]$	-11 $[-27, +6]$	0.279	±10.4
SpaceInvaders, actuator noise (G3)	8	-11.0 $[-13.8, -8.2]$*	-0.5 $[-2.1, +1.0]$	$+1.8$ $[+0.9, +2.7]$*	0.0001	±8.0
Freeway, reward flip (G4)	4	-30 $[-35, -26]$*	-7 $[-12, -3]$*	-2 $[-4, +1]$	0.0007	±3.8
SpaceInvaders, observation noise (G5)	8	-10.6 $[-12.7, -8.5]$*	-0.3 $[-2.1, +1.6]$	$+1.0$ $[-0.5, +2.5]$	$< 10^{-4}$	±8.0
SpaceInvaders, reward extinction (G6)	8	-10.3 $[-12.9, -7.7]$*	$+0.2$ $[-1.1, +1.6]$	-0.4 $[-1.6, +0.9]$	$< 10^{-4}$	±8.0

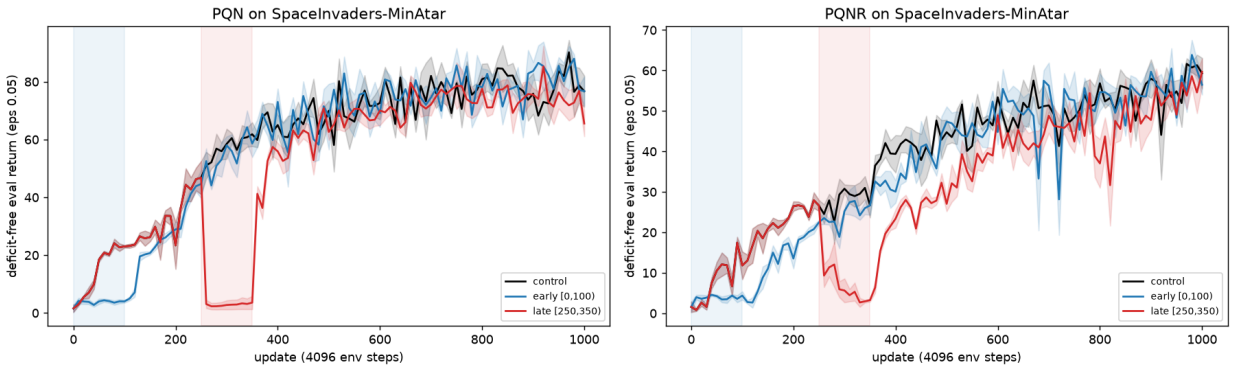


Figure 7: PQN and PQN-R recovery after early and late movement restrictions ($n = 4$).



Figure 8: PPO 2^3 actor × critic × optimizer surgery after the late movement restriction ($n = 4$).

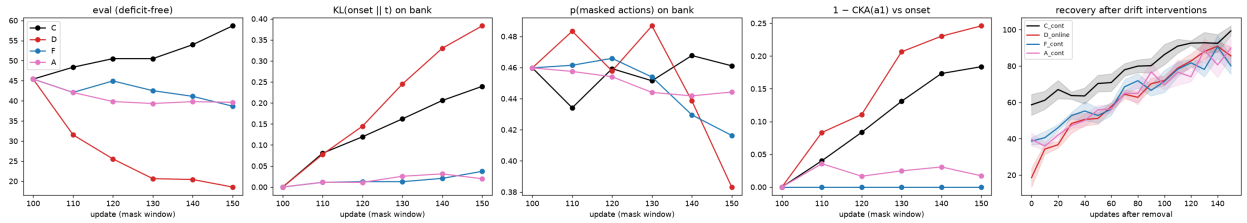


Figure 9: PPO drift suppression during the restriction. F: actor hidden layers frozen; A: KL anchor to the onset policy. Both raise evaluation at removal but leave recovery AUC unchanged.

Table 20: PPO drift-suppression arms at removal and during recovery ($n = 4$; paired differences vs. the unprotected restricted agent D unless stated).

arm	KL from onset	1−CKA (layer a1)	eval at removal	recovery AUC
C (clean)	0.239	0.184	58.7	79.3
D (restricted)	0.383	0.246	18.5	65.0
F (frozen torso)	0.037	0.000	38.7	66.1
A (KL anchor)	0.020	0.017	39.6	64.9
F − D			$+20.1\,[+8.2, +32.0]^*$	$+1.1\,[-3.4, +5.7]$
A − D			$+21.1\,[+13.1, +29.0]^*$	$-0.0\,[-1.8, +1.7]$
F − C			$-20.0\,[-33.5, -6.5]^*$	$-13.2\,[-20.0, -6.4]^*$
A − C			$-19.0\,[-36.0, -2.1]^*$	$-14.4\,[-18.5, -10.3]^*$

Table 21: Post-hoc: minimum detectable effects (MDE, 80% power, two-sided $\alpha = 0.05$, paired t) of the small-n null results, derived from the reported intervals. "Excluded beyond" is the interval's lower bound. Rows marked ‡ were superseded by pre-registered larger-n tests. Source: `p8_power-posthoc.md`.

contrast	n	estimate	MDE	harm excluded beyond
PPO move late, delay-adjusted at +250	4	$+2.0\,[-0.1, +4.1]$	2.7	-0.1
PPO fire late, delay-adjusted at +250	4	$+2.4\,[-3.6, +8.3]$	7.8	-3.6
PQN late, delay-adjusted AUC	4	$-1.4\,[-3.5, +0.7]$	2.7	-3.5
PQN-R late, delay-adjusted at +650	4	$-0.2\,[-7.7, +7.3]$	9.8	-7.7
drift suppression F − D, AUC	4	$+1.1\,[-3.4, +5.7]$	5.9	-3.4
drift suppression A − D, AUC	4	$-0.0\,[-1.8, +1.7]$	2.3	-1.8
PPO critic main effect‡	4	$-2.3\,[-6.2, +1.7]$	5.2	-6.2
PPO optimizer main effect‡	4	$-0.1\,[-2.8, +2.6]$	3.5	-2.8
point-of-no-return pilot L_{300}‡	4	$-1.4\,[-8.2, +5.5]$	9.0	-8.2

Table 22: Post-hoc (exploratory seeds; descriptive): recovery after restrictions under three counterfactual baselines (AUC of restricted − baseline over checkpoints $+10..K$; paired 95% t-intervals; $n = 4$). B1 same-update $C(w_1 + k)$; B2 pause $C(w_1 + k - L)$, the primary estimand E_{damage}; B3 competence-matched $C(t^\star + k)$ (E_{plast}), where $t^\star$ is the first checkpoint at which the clean run's 3-checkpoint moving average reaches the restricted agent's competence at removal. Negative = restricted agent worse. Source: `p8_alt_baselines-posthoc.md`.

agent	arm	K	$t^\star$ (mean)	B1 same-update	B2 pause	B3 competence-matched
PPO	move late $[100, 150)$	250	35	$-10.1\,[-13.0, -7.2]^*$	$+1.1\,[-2.9, +5.2]$	$+18.0\,[+3.8, +32.2]^*$
PPO	fire late $[100, 150)$	250	30	$-15.2\,[-24.8, -5.6]^*$	$-4.0\,[-16.4, +8.4]$	$+14.4\,[+7.0, +21.8]^*$
PQN	late $[250, 350)$	650	12	$-4.6\,[-7.2, -2.0]^*$	$-1.4\,[-3.5, +0.7]$	$+17.5\,[+16.6, +18.3]^*$
PQN-R	late $[250, 350)$	650	25	$-8.6\,[-11.6, -5.6]^*$	$-4.3\,[-7.0, -1.6]^*$	$+7.2\,[+3.5, +11.0]^*$
PPO	move early $[0, 50)$	250	8	$-0.1\,[-15.6, +15.4]$	$+15.0\,[-0.2, +30.1]$	$+12.5\,[-5.1, +30.1]$
PQN	early $[0, 100)$	900	15	$-0.5\,[-4.5, +3.5]$	$+6.4\,[+2.4, +10.5]^*$	$+5.4\,[+1.8, +8.9]^*$
PQN-R	early $[0, 100)$	900	40	$-2.7\,[-5.2, -0.1]^*$	$+2.7\,[-0.2, +5.5]$	$+0.4\,[-2.3, +3.1]$

Table 23: Post-hoc: carrier effects relative to the clean reference agent's recovery AUC, for comparison across algorithms and games (raw AUC scales differ). Relative = mean effect / mean reference AUC. PPO: 150-update recovery; PQN-R: 650 updates. Sources: the analysis files of Tables 6, 8 and 12 and 7.

agent, setting	reference AUC	total	policy / Q-network	critic or buffer	optimizer
PPO, SpaceInvaders movement 2×128 ($n = 12$)	80.6	−15.1 (−18.7%)	actor −13.7 (−17.0%)	critic −1.9 (−2.4%)	+0.2 (+0.2%)
PPO, SpaceInvaders movement 2×256 ($n = 8$)	94.2	−19.7 (−20.9%)	actor −15.7 (−16.7%)	critic −2.5 (−2.7%)	−0.4 (−0.4%)
PPO, SpaceInvaders reward flip ($n = 8$)	79.9	−32.3 (−40.4%)	actor −32.8 (−41.1%)	critic +1.2 (+1.5%)	−1.9 (−2.4%)
PPO, Asterix reward flip ($n = 8$)	6.7	−2.3 (−34%)	actor −2.1 (−31%)	critic −0.1 (−1%)	+0.0 (0%)
PPO, SpaceInvaders reward flip, onset (pause) reference ($n = 8$)	67.4	−19.8 (−29.4%)	actor −22.0 (−32.6%)	critic +2.8 (+4.2%)	−1.6 (−2.4%)
PPO, SpaceInvaders movement, onset (pause) reference ($n = 4$)	67.3	−2.3 (−3.4%)	actor −3.9 (−5.8%)	critic −0.2 (−0.3%)	+1.0 (+1.5%)
PQN-R, 100k buffer ($n = 12$)	48.9	−6.7 (−13.7%)	network −5.6 (−11.5%)	buffer −1.0 (−2.0%)	–
PQN-R, 1M buffer ($n = 8$)	37.9	−5.6 (−14.8%)	network −3.2 (−8.4%)	buffer −2.4 (−6.3%)	–

Table 24: Post-hoc: interaction structure of the PPO 2^3 surgery (movement, 2×128, $n = 12$; AUC over 150 updates, paired 95% t-intervals). Two-way interactions: half the change in one component's effect when the other is damaged, averaged over the third. Relative = effect / clean-agent AUC (80.6). Source: `p10_interactions_move_n12-posthoc.md`.

contrast	AUC	relative
damaged actor into clean agent, DCC − CCC	−13.3 [−16.2, −10.4]*	−16.6%
damaged actor into damaged agent, DDD − CDD	−13.3 [−15.6, −11.0]*	−16.5%
damaged critic into clean agent, CDC − CCC	−0.05 [−1.9, +1.8]	−0.1%
damaged critic into damaged agent, DDD − DCD	−3.0 [−5.3, −0.8]*	−3.8%
damaged Adam into clean agent, CCD − CCC	+1.1 [−0.8, +3.0]	+1.3%
damaged Adam into damaged agent, DDD − DDC	+0.05 [−1.9, +2.0]	+0.1%
actor × critic	−0.5 [−1.9, +0.9]	−0.6%
actor × Adam	+0.5 [−1.0, +2.1]	+0.6%
critic × Adam	−1.0 [−2.1, +0.1]	−1.3%
three-way	+0.4 [−0.4, +1.2]	+0.5%
total − sum of single swaps from CCC	−2.8 [−5.9, +0.4]	−3.5%

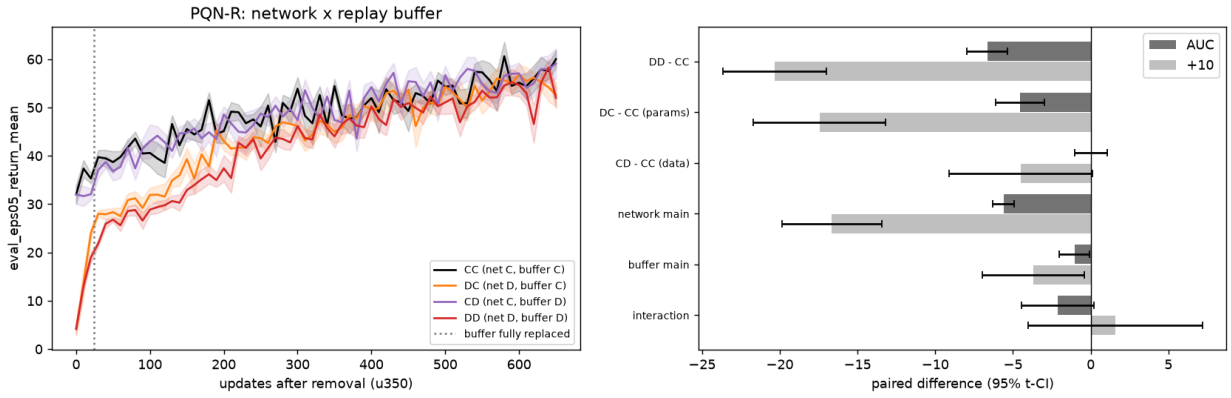


Figure 10: PQN-R network × buffer surgery with a 100k buffer ($n = 12$).

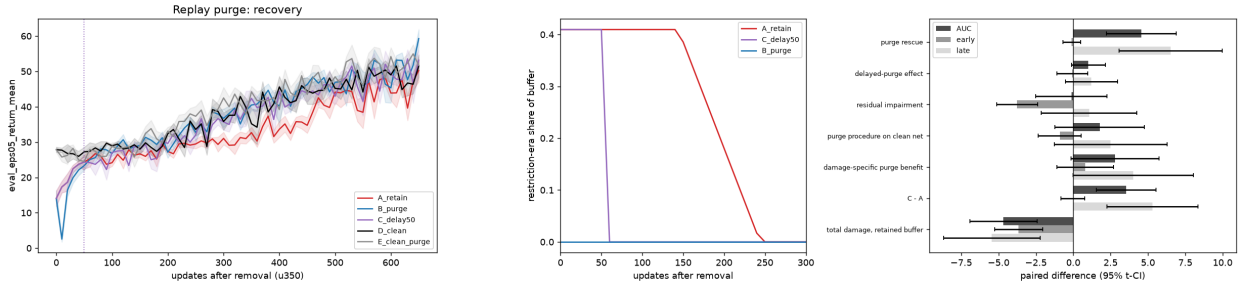


Figure 11: Rung 1: full replay purge ($n = 8$). A: retain; B: purge at removal; C: purge delayed by 50 updates; D: clean agent; E: clean agent with purge.

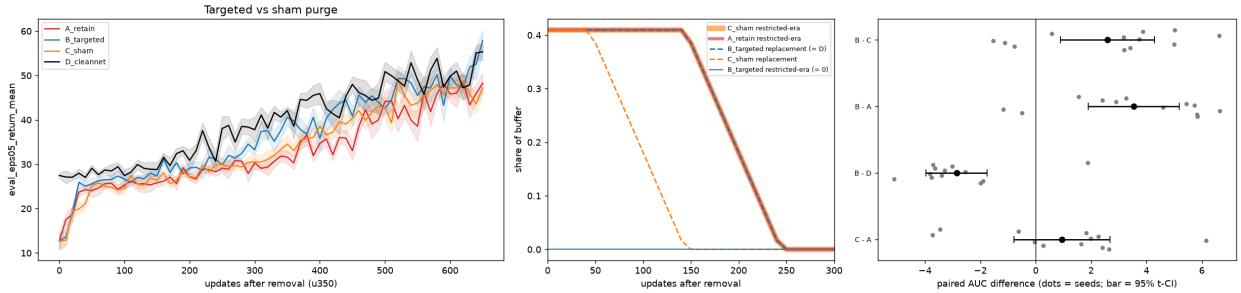


Figure 12: Rung 2: targeted purge of restriction-era rows vs. an age-nearest sham ($n = 12$).

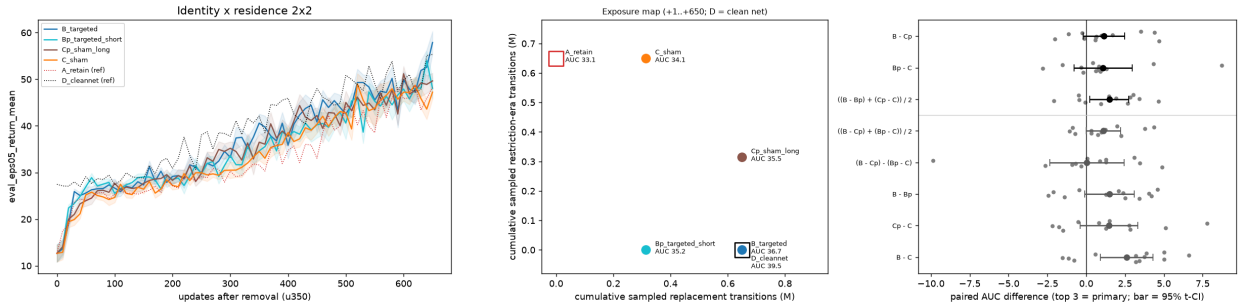


Figure 13: Rung 3: identity $\times$ residence 2×2 design ($n = 12$).

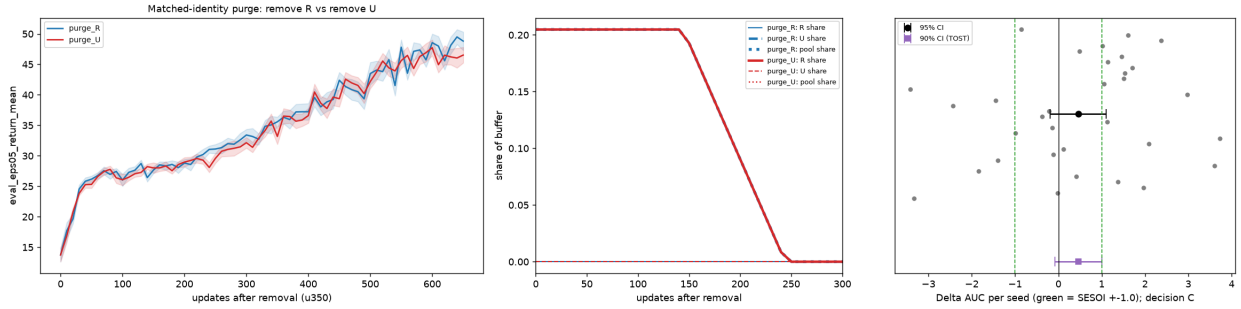


Figure 14: Rung 4: matched-identity purge ($n = 32$) with the pre-registered equivalence margin.

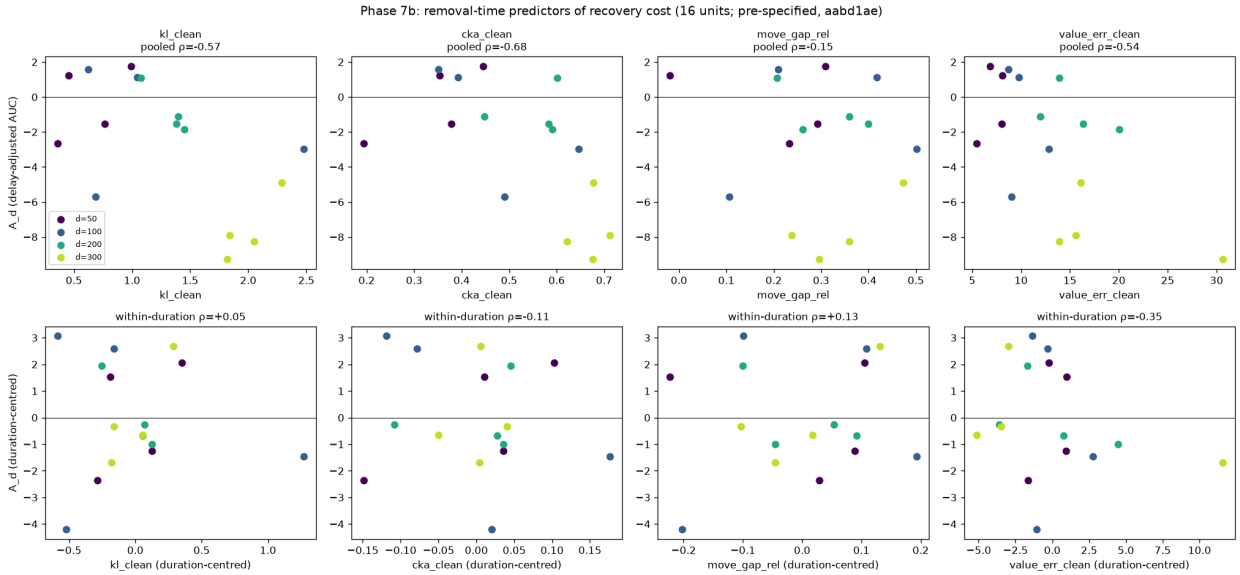


Figure 15: Phase 7b: pre-specified removal-time predictors against recovery cost. Pooled associations follow restriction length; within-duration associations were not detected ($n = 4$ per duration).